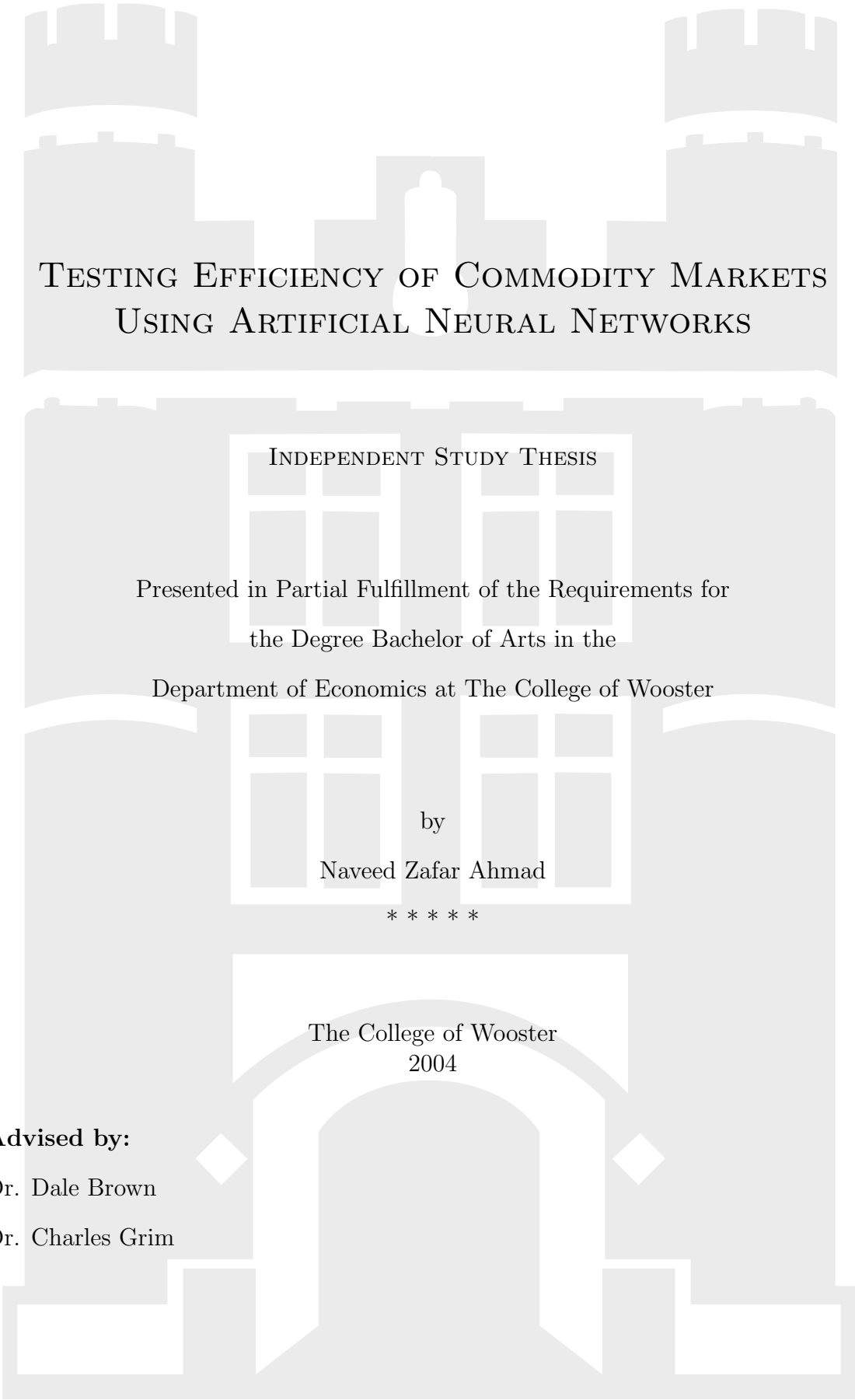




TESTING EFFICIENCY OF COMMODITY MARKETS USING ARTIFICIAL NEURAL NETWORKS

INDEPENDENT STUDY THESIS

Presented in Partial Fulfillment of the Requirements for
the Degree Bachelor of Arts in the
Department of Economics at The College of Wooster

by

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* * * * *

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Abstract

This study used feed-forward neural networks using back-propagation learning algorithm to test efficiency of the soybean futures commodity market. The soybean futures market was tested for the weak form of efficiency outlined by the Efficient Market Hypothesis first proposed by Fama (1970). The study found that the soybean futures market to be efficient in the weak form since positive profits were not realized after applying a trading rule to the predictions from the neural network.

Dedicated to my family, especially my parents!

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Introduction

"Individuals" trade in the commodity markets to hedge against potential risk of moving spot prices. They take positions to ensure a certain level of prices of the commodity at a future date. These individuals make use of derivatives such as futures contracts, a standardized form of a forward contract to hedge for their position. These contracts are traded in commodity exchange markets. Futures contract prices reflect the price movement in the spot market. Efficiency in the futures market ensures that the futures prices are the best predictors of the future spot prices.

The purpose of this study is test if the soybean futures commodity market is weak form efficient as the definition presented in the efficient market hypothesis first presented by Fama (1970). The study hypothesizes that the soybean futures commodity market is not weak form efficient and that consistent profits can be made using neural networks by efficiently processing easily available information for soybean price indicators.

The efficiency of the soybean futures commodity market is tested through a feed forward artificial neural network using back-propagation leaning algorithm. These networks as the name implies are artificial representations of the neural networks

present in the human brain. These networks have evolved from a discipline of computer science known as Artificial Intelligence (AI). The goal of AI is to develop computer based systems that mimic human behaviors.

Neural networks built on back-propagation algorithm have the capability of finding accurate solutions in an environment characterized by complex, noisy, irrelevant, or partial information. This capability of the neural networks using back-propagation algorithm will be tested in this study. Approximately thirty five soybean spot price indicators will be used to determine the soybean price in the future.

CHAPTER 1

Theory

1.1 Commodity Markets

In the US, commodity markets arose in the late 1700s. The markets were primarily cash markets for spot delivery. Producers at the time were concerned with demand and supply related price fluctuations. The influx of goods at a harvest time overwhelmed the short-term demand for the goods, and the market found itself unable to absorb the glut at any reasonable price. Thus, these goods were often sold at extremely low prices due to lack of demand and storage facilities. The quantity, as well as the quality, of the infrastructure in the US were important factors. During most of the year, weather caused roads and railroads to shut down; underdeveloped harbor facilities were also a hurdle. Thus producers were unable to get the goods to market in a timely manner, and faced inadequate demand. The speculators of the early commodity markets made inroads with the legislative bodies to improve road, railway, and waterway systems to ease the transportation of goods to the cities (CBT 3).

By the late 1840s, merchants had formed formal commodity trading exchanges

in many cities. The Chicago Board of Trade (CBT), established in 1848, was one of the first such commodity exchanges. Forward contracts were the main derivative contracts traded in this facility. Early forward contracts solved the problem of supply and demand by matching sellers with buyer prior to the harvesting time, but didn't address such matters as risk involved with low yield, transportation, quality, and price changes. The buyers and the seller were matched up in the the exchanges and contracts were written up to the agreements of both the parties. The forward contracts were not standardized therefore the price, quality, quantity, time and day of the transaction was different for each contract and these variables were agreed upon by both the seller and buyer. By the late 1800s, the commodity markets evolved, formalizing trading practices, standardizing contracts, establishing settlement procedures, and developing methods for determining the market clearing price. These markets eventually changed the informal forward agreements to such modern day derivative contracts, such as the futures contract. The markets also set specifications for commodities such as grades. The futures contract were developed to address the transportation costs, and they set quality standards for the commodities that were traded with these contracts. The standardization of quality, quantity, time and place of transaction made the trading more efficient thus increased trading in these contracts (CBT 8).

Twelve major exchanges around the US, currently allow trading in many kinds of different derivative contracts and of different commodities such as grain, livestock, oil, meal, metal, minerals, forest products, textiles, foodstuffs, and foreign currencies. These commodity exchanges have open trading sessions, much like the stock

exchanges. The Central Exchange Authority (CEA) regulates the contracts. This authority is also responsible for setting up commodity exchange locations and quality control.

1.2 Commodity Exchanges

Commodity exchanges came about to provide an organized and regulated facility for conducting the business of trading commodities. Unlike the stock exchanges, where a piece of the ownership of a firm is traded, in a commodity exchange only derivatives are typically traded. A few exchanges like the Minneapolis Grain exchange will actually facilitate the physical transfer of commodities, but most exchanges have nothing to do with the physical commodities.***

Exchanges are private associations incorporated in the states in which they reside. They consist of individuals who gather in order to trade in commodities. Usually only members may trade in these facilities. However, in some cases such as the MidAmerica Commodity Exchange, the floor is open to the public. Membership in exchanges is limited and seats sell by auction when they become available. Therefore, investors and speculators work through brokers or companies who have links to members on the exchange to trade in contracts. Larger brokerage firms may have several seats in different commodity exchanges, and they can underwrite their own employees to represent these firms on the exchange floors. The numbers of members may change over the years, depending on the business a particular exchange attracts. Each of

the commodity exchanges has its own criteria for membership; they may include individual background, financial status, and company affiliation(Gould, p.64).

The main work of an exchange is to formulate and supervise trading of the different contracts on its floors. It regulates the trading of members on the floor, and settles disputes between its members. It monitors trading activity and gets recent news to the traders in a fast and unbiased manner. If required, the exchange provides means for the traders to liquidate their positions or settle them.

Commodity exchanges provide only the resources necessary to trade contracts, and in some cases, will devise new contracts to attract more business. The exchange is not supposed to interfere with commodity prices since these are determined through the market forces of supply and demand. The exchange does not trade any contracts for its own benefit. It makes its money through dues, fees, and the sale of information. The success of an exchange depends upon the volume of contracts traded on its floor. Thus the profits depend upon the members, administration, and employees of the exchange.

1.3 Contracts

In the futures commodity markets futures contracts and options on these futures contracts are traded. The futures contracts have their origins in the forward contracts. The futures contracts are essentially standardized forward contracts developed for the purpose of decreasing transaction costs and increasing liquidity of the contracts. Liquidity is defined as the relative speed and ease an asset can be converted into

cash. In case of futures contract because of their standardization only the price of the contract must be negotiated between the seller and the buyer. The development of this price through the supply and demand forces of the traders, and the relation of the futures price to other factors such as cost of carry and the spot price of the underlying will be shown later in this section.

1.3.1 Forward Contract

The forward contract is between two parties for a delivery of a certain quality and quantity of goods at a particular time and place in the future at a negotiated price. The dates settled in a forward contract are the delivery date, settlement date, and the maturity date. The parties involved in this deal are called counterparties. There is no exchange of any underlying asset at the time the contract is forged, unless one of the parties demands "good faith" money to serve as collateral for the terms in the contract. Typically an agreement is made between the two counter parties to transfer ownership of the asset mentioned in the contract at the price quoted on a future date.

A forward delivery contract might specify that a delivery of 10,000 bushels of corn one year from today at \$2 per bushel, from the seller of the contract to the buyer of the contract. If the price of the corn bushel stays at two dollars per bushel then the contract has no intrinsic value to either party, thus no cash or commodity exchanges at the maturity date of the contract. If the spot price of the corn increases to \$3 per bushel one year from now, then the long position makes a profit equal to \$10,000. As the underlying commodity in the contract is worth \$1 more than the price the buyer

has to pay the seller. On the other hand if the spot price of the corn decreases to \$1 per bushel then the short position makes a profit of \$10,000 and the long position loses the same amount as he or she would have preferred to buy the corn in the spot market for a cheaper price.

The profits and losses are realized at delivery for both counter parties. Forward contracts can be involve either physical settlement at maturity or cash settlement. In case of a cash settlement, the long receives a cash payment if the price of the commodity increased. Otherwise the short receives a cash payment from the long position. In a physical transaction the long position, which could be a farmer delivers the commodity agreed upon in the contract to the time and location also settled in the contract. It is important that the deliverer have the quantity and quality settled by the contract.

Forward contracts play an important role not only as financial instruments, but many other financial instruments with more complex features can be described using many different long and short positioned forward contracts as can be seen in the next section.

1.3.2 Future Contracts

A futures contract is essentially a forward contract trading on an organized exchange with key institutional features aimed at overcoming some of the problems faced by traders. These problems include credit risk exposure, the difficulty locating a trading partner, and the need for an efficient means of termination of contract prior

to its expiration. The organized exchanges have a framework of rules and regulations in place to cope with these deficiencies in the forward contracts.

Credit risk is exposure to the possibility of losing money. In case of commodities the credit risk can involve the spot price changing to an unfavorable position. For a trader in the long position which can be the farmer, it can be an increase in the price of the underlying commodity listed in the forward contract. Since the farmer is obligated to come true on the contract therefore he will lose the profit that he could have made had he sold the commodity in the open market at the spot price. For a farmer credit risk can also be a problem if he/she does not have the quality or the quantity of the commodity agreed in the contract. On the other hand the credit risk for a short position or the buyer of the commodity can be a decrease in the spot price of the commodity listed in the forward contract. The buyer will incur a loss equal to the decrease in the price of the commodity because he is obligated to buy the commodity from the farmer in the quantity and price agreed upon in the contract. The buyer is also open to credit risk if in the long position in this case the farmer does not deliver the commodity as agreed upon in the contract.

To overcome the problem of credit risk, futures exchanges require daily adjustment of gains and losses. This realization of profits and losses is called 'marking the market'. The trader who has profited from the day's trade can withdraw the amount over the value from the last mark-to-market period. This marking to the market creates a difference in the way futures and forwards allow traders to lock in the prices. In case of a forward contract the price paid at delivery is the price stated on the contract. With the futures contract, the buyer pays and the seller receives the spot

price prevailing at the delivery date. Hence, the price is locked in through the profits and losses realized through the life of the contract. This means that gains and losses on a futures position, recognized daily in conjunction with the spot price at delivery yield a net price corresponding to the futures price quoted at the time the future position was established (Overdahl and Overdahl, p. 27).

The procedure of marking to market requires that the trader open a margin account with a clearing house connected with the organized exchange. The margin account holds either cash or short term securities such as T-Bills which are enough to cover the maximum daily loss on their futures position. The reason short term securities are allowed as margin payments because the traders would lose out on the interest they could have made on the cash that had to be kept as margin payments. Traders who fail to meet their margin level have their position liquidated by the exchange before trading resumes the following day.

The exchange imposes a daily price limit on the futures contracts both a ceiling and a floor to define the maximum loss. If the limit is hit, then trading is halted for the day, and can resume the next day at a price within the range of the next days price limit. Therefore marking to market and the imposition of a price ceiling and floor serves to reduce exposure of extreme financial loss to the traders in the organized exchanges. The price floors and ceiling while solving one problem create another. If the spot price of the underlying commodity in the contract increases more rapidly than the floors and ceilings then there could be a potential problem of price adjustment of the futures contract to accurately reflect the spot price of the underlying commodity.

The second problem that organized exchanges address is the diversity of forward contract terms, which make it difficult for traders to find counter parties. Forward contracts have terms customized to complement the individual needs of a particular party. A counter party with the same unique needs and an opposite position on the underlying asset may be hard to find. To overcome this problem organized exchanges offer standardized contracts and centralized trading facilities, therefore enabling traders to meet and be able to agree on the trade easily. This also economizes the cost of searching for a partner for the traders.

The problem of exiting a position, short of actually completing the delivery is also solved by the organized exchanges through the working of the clearing houses. There are two options for each trader to fill his obligation described by the contract. He can either deliver the goods on the said date, or he can negotiate with the counter party to terminate the contract early, a process that typically requires a payment in the form of cash by the trader to the counter party. The term as well as the physical mitigation between the two parties is carried out by the clearing house. Most futures contracts are closed without delivery. The counter parties exchange the cash and then if one of the parties wishes to buy or sell the commodity they can do so in the spot market and the profit from the contract can compensate them for any losses they might incur. In some futures contract delivery is not even possible, such as the futures contract for the S&P 500 stock index. Today, all the traders interact with the clearing house. This promotes anonymity between the traders trading the contracts, thus increases liquidity in the market since all traders have to deal with only one

entity, the clearing house. In this way, some of the risk is absorbed by the clearing house.

Standardization of contracts gives rise to the problem of quality assessment of the grain if the holder chooses to accept delivery. This is solved by an introduction of premium or discount. Through specifying in the contract the different grades that are deliverable, and by establishing set premiums and set discounts, producers and buyers have considerable flexibility in their use of futures trading. As mentioned above positions can be cash settled where the physical commodity does not exchange hands. By keeping homogeneity in contracts the futures market adds liquidity for all traders.

1.3.3 Future Contract Price Quotations

After a day of trading at the Commodity Exchange, the price of the futures contract for soybean is readjusted according to the spot price and expected future spot price at the contract maturity. The price quotes for futures contracts on different commodities are available on the internet with a ten minute delay. Real time quotes can be obtained from specific vendors on a subscription basis. Futures quotes are also published each day in the *Wall Street Journal* and other newspapers. The quotes are reported in the manner represented by table 1.1.

Illustration of the futures price quotes for soybeans can be seen in the table 1.1 for November 6th, 2003. Although there are many different futures contracts for each commodity, they all have the same underlying features. Referring to the first column in table 1.1, the symbol SXO3 provides three pieces of information. The first letter ‘S’

corresponds to the commodity soybeans. The second letter 'X' represents the month of expiration for the contracts(in this case November), and the last two digits '03' stand for the year. The next three columns in the table are self-explanatory. The month represents the month of expiration for the contract, while the date and time provide information pertaining to the date and time of the last traded contract of the particular expiration date. The open price is the price that the contracts started trading at when trading started in the morning. The High and Low prices are the highest and the lowest price the contract was traded during the day, and the last price is the closing price of the day for the contract in question. The change is the difference between the last day's closing price and current day's closing price.

Interpreting the price of a commodities contract is different for each commodity. The price is dependent upon the information in the contract. The simple soybean contract represented in the table above is for 5000 bushels. The price is quoted in cents per bushel. Thus each cent is equal to \$50. So the opening price of SXO3 contract would be \$38,500 ($770 \times 50 = \$38,500$). The last digit in the price quote represents the minimum fluctuation allowed by the contract which is a quarter of a cent (\$12.50). The increments or ticks are in increments of two for each quarter of a cent. Thus, the change for SXO3 contract is calculated as

$$-7 \dots 7 \times 50 = -\$450$$

$$'4 \dots 4 \div 8 = 0.5 \times 50 = \$25$$

Thus the total change in the dollar amount would be -\$475. The letter 's' at the end

SOYBEANS (DAY) TABLE 1.0										
Symbol	Month	Date	Time ^a	Open	High	Low	Last	Chg	Volume	OpenInt
SX03	Nov '03	11/06/03	13:39:49	770'0	771'4	757'0	757'6s	-7'4	8316	5784
SF04	Jan '04	11/06/03	13:39:49	778'0	779'0	764'0	765'0s	-8'0	72294	129752
SH04	Mar '04	11/06/03	13:39:49	771'0	773'0	758'4	759'6s	-6'2	10916	38887
SK04	May '04	11/06/03	13:39:49	737'0	740'0	731'0	732'0s	0'0	5323	38580
SN04	Jul '04	11/06/03	13:39:49	724'0	729'0	720'0	722'6s	2'2	3924	18656
SQ04	Aug '04	11/06/03	13:39:49	705'0	707'0	700'4	701'0s	4'0	259	2414
SU04	Sep '04	11/06/03	13:39:49	658'0	660'0	653'2	653'2s	2'0	53	574
SX04	Nov '04	11/06/03	13:39:49	596'0	598'0	590'0	593'6s	1'6	2093	9099
SF05	Jan '05	11/06/03	13:39:49	600'0	600'0	597'0	597'0s	8'0	2	5
SH05	Mar '05	11/06/03	13:39:49		589'0	589'0	589'0s	2'0	3	7
SN05	Jul '05	07/18/03	08:59:50							
SX05	Nov '05	07/12/03	13:38:20				539'4			

Table 1.1: Soybean Futures Contract's Price Quotations

of the contract prices in column labeled 'Last' indicates that the price listed as the last was the settlement price of the soybean commodity contract for the day.

The volume stands for the total number of trades made for the particular contract that day. That means that different SXO3 contracts exchange hands 8316 times. The open interest number stands for the total number of contracts outstanding at the end of the trading day. Put another way, it is the number of contracts currently obligated for delivery. If a buyer and a seller trade one contract, and they don't make a reversing trade then the open interest increases by one. Reversing trade means that trader A and trader B trade a contract and trader A turns around and trades a contract with trader C in the opposite position. So in affect trader C has simply taken the place of

trader A. When a reverse trade is made by one of the parties involved then the open interest stays the same. Otherwise if trader B also make a reversing trade then the number of open interest goes down.

All contracts begin with zero open interest, and end with zero open interest. When trade opens for a particular contract there is no open interest, and after the contract expires no interest is left open as they have all been settled. Figure 1.1 summarizes the concept of open interest. As time increases after the contract is permitted to be traded the open interest grows at a slower pace until the time to maturity is near.

As time to maturity nears the contracts start to be settled until the contract expires and no more open interests remain. A futures contract becomes available for trading several months before it expires. Generally in the case of soybean commodity futures, the contract is opened for trading at least a year before its expiration. There are about seven soybean futures contracts that expire in each year. So at any given time there are more than seven contracts open for trading in soybeans. Generally not many people start trading in contracts as soon as it is available for trading. As the time for the expiration of the contract becomes nearer more people are willing to trade in the contract because the current market condition affects the expiration of the contract price more. Therefore it is easier to predict where the price of the contract will lead, therefore making it more attractive for traders. Buyers and sellers of soybeans are more interested in covering their losses for the short run which is within the current year, rather than the next year. Therefore the open interest increases for a contract though about mid way of its trading life, as the graph above indicates.

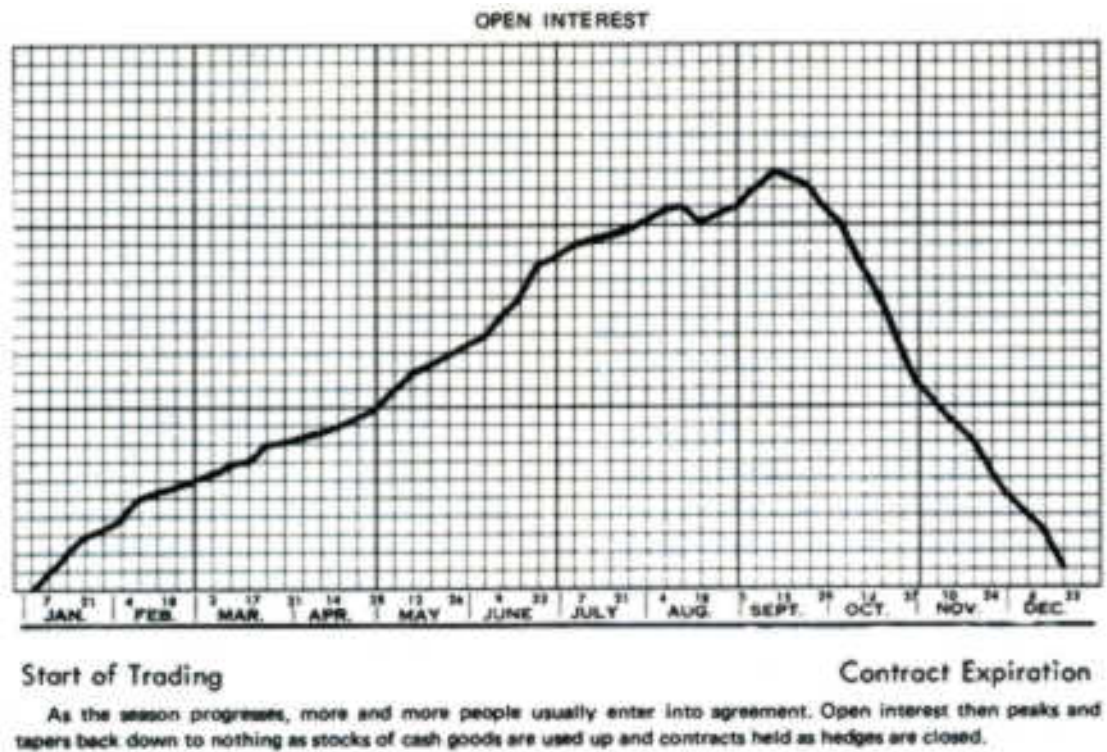


Figure 1.1: Graph of Open Interest for a Contract through it's life span

As the contract gets near its expiration date most of the position are closed, and as mention earlier most of the contracts are cash settled. Very few contracts are held to expiration where the delivery on the contract is made. Most traders prefer to make their gain from hedging the market and buying their respective commodities from closer location in order to reduce transportation costs.

1.3.4 Future Contract Pricing

The futures contract price is determined daily through the supply and demand of the traders involved in trading these contracts. The supply and demand of these contracts depends on the expected price movement of the spot price of the underlying commodity in the future. It can be stated the futures contract price represents the expected spot price of the underlying commodity at the time of the expiration of the contract. The condition that the futures contract price will be equal to the spot price at expiration is explained through the arbitrage based model known as the Cost-of-Carry model. For understanding the basics of this model, the perfect market scenario will be represented first, and then modified to fit the imperfect (real) markets. The behavior of the futures contract prices and the current spot prices are different depending on the type commodity: seasonal and non-seasonal. The commodity of interest for this study is soybeans, which is a seasonal commodity. The analysis of the pricing of the futures contract will be done in relation to seasonal commodities.

To analyze the pricing of the futures contract an analysis of profit and loss through buying and selling futures and spot market is necessary. Thus, let $s(t)$ be the spot price of the commodity at time t , and let $f(t,u)$ be the price of the futures contract at time t expiring at time u . A trader in the futures market who buys a contract at time 0 and sells it at time 1, will make a profit equal to $f(1,u) - f(0,u)$. Another trader who sells a future contract at time 0 and buys a contract at time 1, will make a profit equal to $f(0,u) - f(1,u)$. In the spot market a trader who buys one unit of a commodity at time 0 and sell the unit in the same market at time 1 will make a

profit of $s(1) - s(0) - z(0,1)$, where $z(0,1)$ is equal to the net carrying cost of the unit from time 0 to 1. The net carrying cost is made up of four components which are listed below:

- Storage Cost: money paid for the storage of the unit of commodity
- Financing Cost: if money was borrowed to buy this commodity, then the financing cost will be equal to the interest paid on the financed capital
- Insurance cost: The money paid to insurance companies to insure the commodity against fire, theft, and similar hazards
- The percentage yield on the value of the amount held: This applies in the case of financial instruments such as bonds.

From the above components the first one depends on the weight, size, and type of commodity while the others depend on the price of the commodity. So this means that a profit will only be made by the trader trading in the spot market if the spot price rises more than the net carrying cost of the commodity from time 0 to 1 (Houthakker and Williamson, p. 256).

Now let suppose that a trader buys a contract and takes delivery on the futures commodity contract when it matures. Suppose the contract was held from time 0 to 1, then the profit made by the trader will be equal to $s(1) - f(0,1)$. This means that the trader will make a profit equal to the futures price at time zero subtracted from the spot price of the commodity at maturity of the contract. Suppose that another trader hedges in the futures market since he bought and holds an inventory

of a commodity from time 0 to 1. The futures contract bought by this farmer is of maturity date u . The profit that will be made by this trader will be:

$$s(1) - s(0) - z(0, u) + f(0, u) - f(1, u) \quad (1.1)$$

Later in this section a special condition will be shown through the cost-of-carry model will provides support that the price of a futures contract at expiration will be equal to the spot price at the time, $f(1, 1) = s(1)$. If such is the case the equation 1.1 can be written as:

$$f(0, u) - s(0) - z(0, 1) \quad (1.2)$$

The difference between the spot price $s(t)$ and the futures price $f(t, u)$ is known as the basis. An illustration of the basis is given below.

Evaluation of the basis is important because it holds information pertaining to the price movement of the futures prices. According to Purcell (1991), the basis level at harvest relative to projected later basis levels and the related futures pricing opportunities provide the keys to profitable hold-sell or storage decisions. Therefore a good prediction of the future level of the basis can lead to better forecasts of the futures contract prices. One of the main factors that effect the basis in seasonal commodities such as grains is the level of inventory. Figure 1.3 shows the path of the inventory accumulation and spot prices for a seasonal commodity over the crop year. For the crop represented in figure 1.3, the harvest begins to arrive in early June, and the supplies keep arriving though July and August. Inventory for any season commodity is built up according to the expected consumption of the next year until the new crop is harvested(Houthakker and Williamson, p. 265).

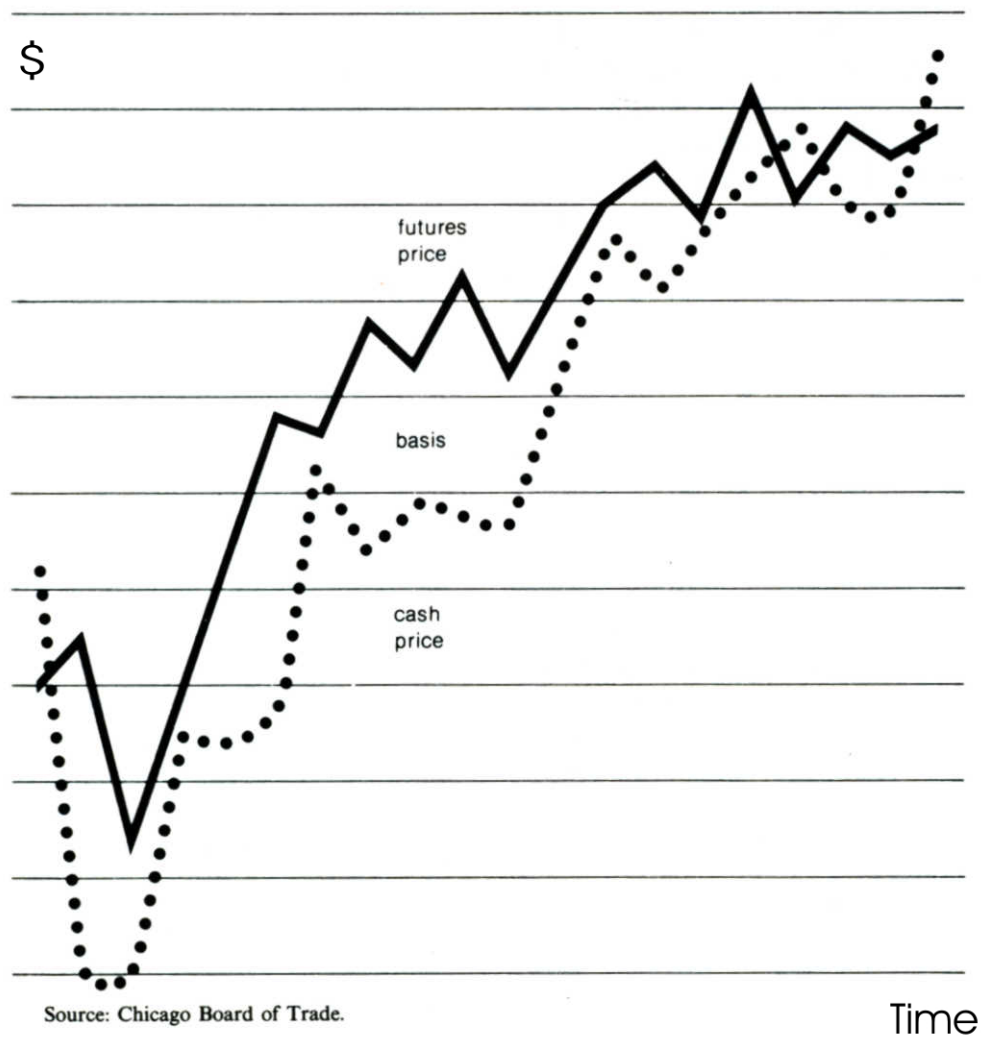


Figure 1.2: Illustration of the basis(Gould, p. 279)

In the case of seasonal commodities the spot price of the commodity begins to decline as the inventory from the new harvest begins to arrive. “Beyond some point

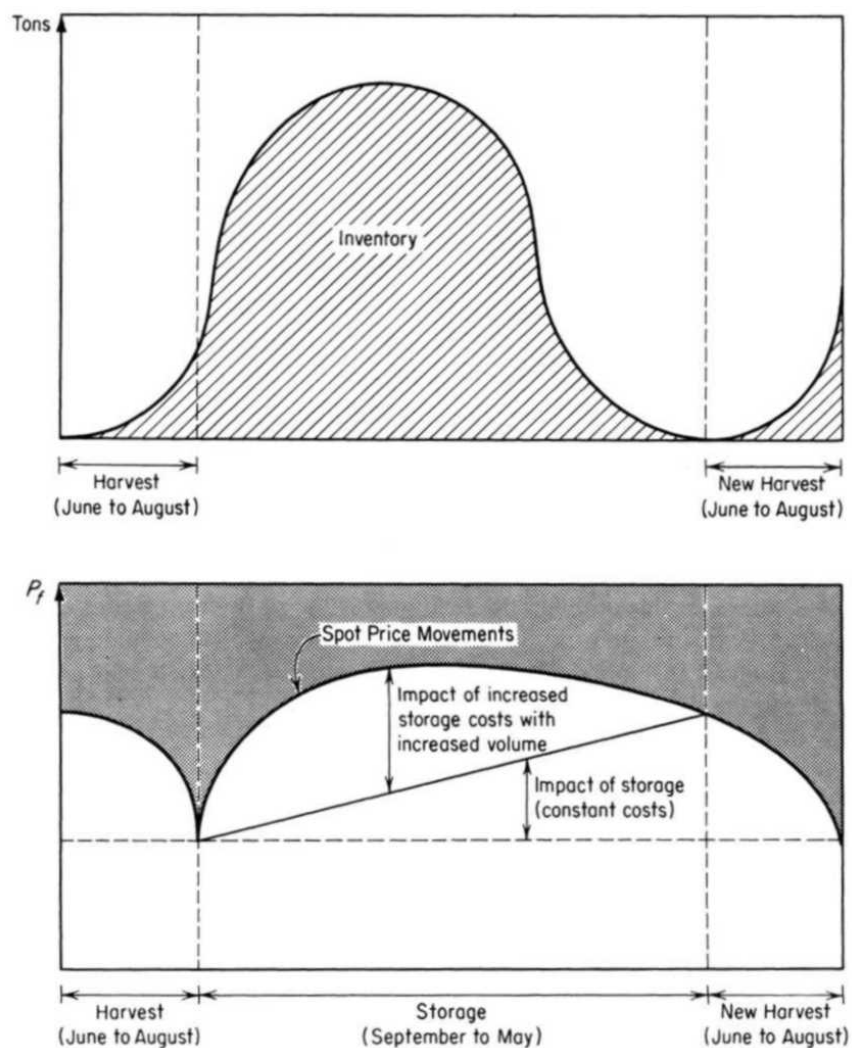


Figure 1.3: Inventory and spot price of a seasonal commodity (Houthakker and Williamson, p. 266)

the spot price is expected to rise over the remainder of the year, reflecting the cost of storing the commodity” (Houthakker and Williamson, p. 265). In figure 1.3 this

point is market at the beginning of September. If the carrying cost of storing the commodity were constant regardless of the total amount stored then a linear rise in spot prices of the commodity would be expected. For example, if carry on unit of soybeans for one day was κ . The cost of carrying the next unit will also be κ since the cost of carrying the commodity is constant. Thus the cost of storing 1000 units of soybeans would be equal to $1000 \times \kappa$. This linear relation is illustrated in the price graph in figure 1.3. However, in reality the cost per unit of storing a commodity increases as the total volume stored increases, either “because of higher operating costs or because inferior methods cause greater loss and deterioration to the commodity during storage”(Houthakker and Williamson, p. 265). The relationship between carrying cost and inventory stored is non-linear. Thus with a unit rise in inventory there will be more than a unit rise in the carrying cost. The non-linear relationship of carrying cost to inventory can be seen in the figure 1.3 as the spot price in relation to inventory is also non-linear because as mentioned earlier the spot price incorporates the carrying cost of the commodity(Houthakker and Williamson, p. 266).

The behavior of the spot prices across a crop cycle affects the futures prices as well. Suppose a March futures contract to buy or sell a commodity will be affected by the expected equilibrium price of that commodity in March. The expected equilibrium price is determined through the expected supply and demand of the commodity in the future. The expected supply of the commodity is determined by the current harvest, current level of inventory, and the expected future harvest. The demand of the commodity is determined through the expected demand of the manufacturers using

that commodity as their inputs. Thus, “seasonality of crops have an important effect on the relationship between the prices of futures contracts with different maturities (e.g, the price of June contract relative to a March contract) and on the behavior of the basis over time”(Houthakker and Williamson, p. 266). Figure 1.4 shows the behavior of the basis of a March crop contract where the crop cycle involves harvest beginning each September.

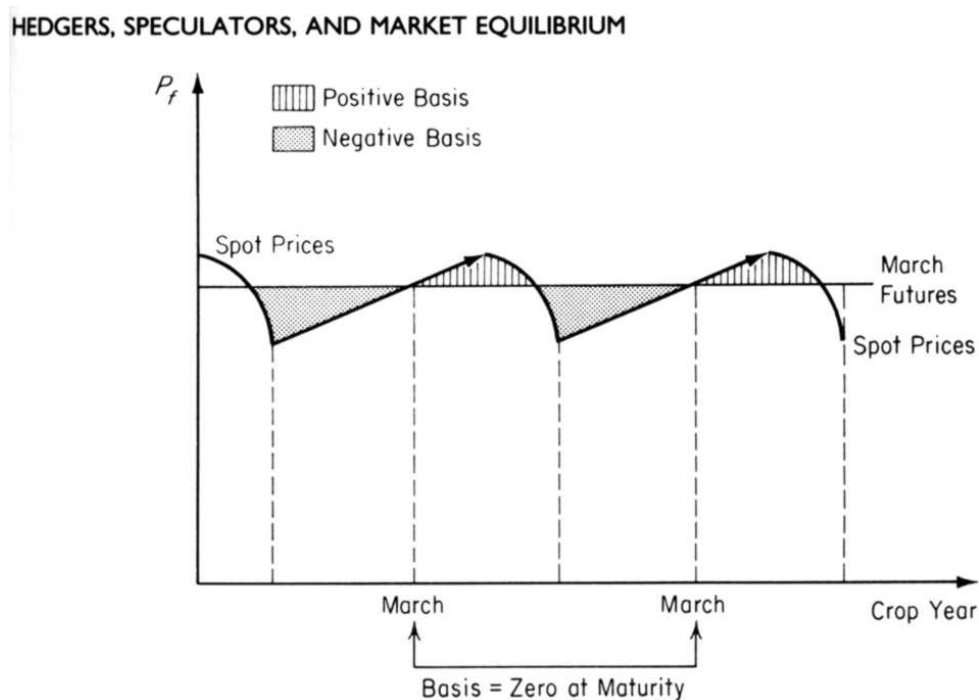


Figure 1.4: Futures prices and the “basis” in a seasonal market(Houthakker and Williamson, p. 267)

In figure 1.4 notice the shift between the backwardation and contango. Backwardation is referred to the instance where the futures contract price approaches the spot from below, meaning the futures price is lower than the spot price. This is shown as the positive basis. Contango is referred to the situation where the contract price is higher than the spot price thus the futures price approaches the spot price from above. This is shown as the negative basis in figure 1.4. Also note that the price of the futures contract is equal to the spot price at the time when the contract expires. The reasoning behind this phenomenon will be presented by the Cost of Carry Model later in this section.

The seasonality of crops also influences the price relationship between contracts of different maturity dates. “Even where supply and demand are the same from year to year, in some months “near” contracts will be more valuable than distant ones” (Houthakker and Williamson, p. 267). This relation can be opposite at other times of the year as the figure 1.5 illustrates. As can be seen from the figure 1.5 that July and March are more valuable than their next near contracts September and July respectively. The March contract on the other hand is more valuable than the September contract. As mentioned earlier the spot price moves in relation to the level of inventory. In September the prices are the lowest as figure 1.3 illustrates. Therefore the September contract is the lower in price than the March contract where the spot prices are considerably higher.

As demonstrated earlier the level of the basis defines the relationship between the futures prices and the spot price. A closer look at the relation between the inventory

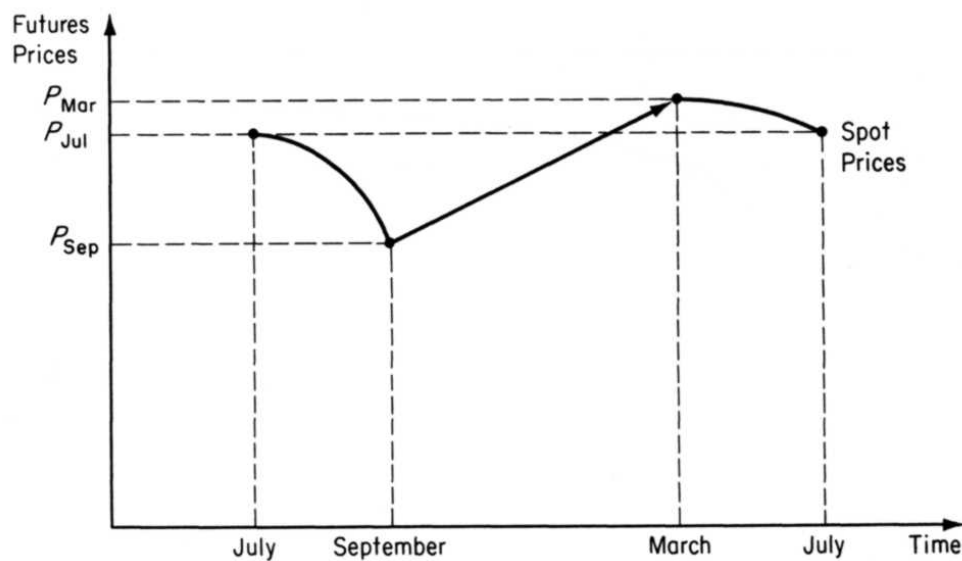


Figure 1.5: Prices of March, July, and September futures contracts for a seasonal commodity with July-August harvest (Houthakker and Williamson, p. 267)

and the basis is in order. Figure 1.6[†] relates the level of inventories to the basis. Even if the inventories are large, still there is incentive to hold them, and the basis is equal to the gross carrying cost which as mentioned earlier rises as more expensive storage space is used. As inventories are depleted the basis takes a negative number since the expected futures price will be much higher in anticipation that, at current consumption levels, the spot price of the commodity will be much higher in the future.

Therefore the relationship between the spot price and the futures price of the

[†]The measurement of the quantities on the axis are relative, although a better measurement might have been in percentage terms.

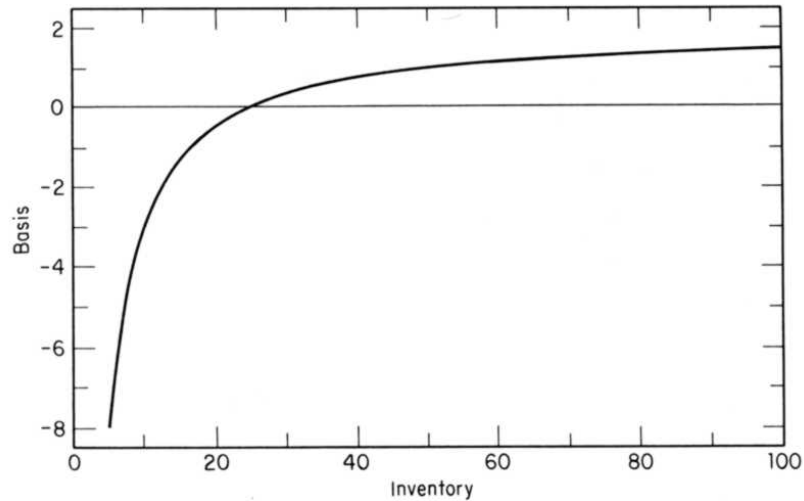


Figure 1.6: The supply curve of storage(Houthakker and Williamson, p. 267)

commodity can be defined through the analysis of the basis. The basis includes the net carrying cost of goods. One of the major components of the net carrying cost is the cost of storage. Thus the level of inventories are closely related to the level of basis. The relationship of the spot prices and the futures contract price at the time of the expiration of the futures contract is important and is explain by the cost-of-carry model.

1.3.4.1 Cost-of-Carry Model in Perfect Markets

Cost of Carry model is used to determine the price relationship between the spot or cash price of commodities and the futures contract price at the time of contract expiration. To explain this model we will use the soybean as the commodity of

interest. This model uses arbitrage opportunities as its bases for price determination. Arbitrage is defined as the attempt to profit by exploiting price differences of identical or similar assets in different markets or different forms. Usually in the case of an arbitrage opportunity, the arbitrageur occurs the least amount of risk possible in the transactions made. So for an arbitrage to take place the profit should almost be certain.

There are four underlying assumptions made by the cost of carry model. There are no transactions costs, carrying costs, restrictions of short selling, unequal borrowing and lending rates, and lack of storability. These assumption do not invalidate this framework in the real world, rather it is able to map the real world markets with a few adjustments.

In the case of perfect markets we will assume that there are no transportation costs. We will assume that the most important charge is the financing charge. Most of the players in the market faces a financing charge, which is equivalent on a short-term basis to that of the interest rates on the repurchase agreements (repo).

Considering only the financing charge the illustration of the cash-and-carry arbitrage is represented in the Table 1.2. In the case presented, there are no restrictions on the use of the proceeds from the short sales. The following is called a cash and carry arbitrage since the trader buys the goods on loan and carries them to the expiration date stated in the futures contract. The trade is an arbitrage because the trader is guaranteed a riskless profit when the transaction is made at time $t=0$. It is to be noted that there is no investment necessary on the part of the trader in this situation. The trader at time $t=0$ will borrow a thousand dollars at five percent

interest rate. He will then use the loan money to buy amount of commodity C, and sell a futures contract for eleven hundred dollars for delivery of amount of C one year from now. At $t=1$, the trader will retrieve the commodity C from storage and deliver as per the contract terms. He will repay the loan plus interest from the proceeds of the transaction above with a profit of fifty dollars (Overdahl and Overdahl, p. 36).

Table 2.0 Cash-Carry Arbitrage Transaction (Overdahl 36)	
Prices for analysis:	
Spot price of commodity C per α amount	\$1,000
Future price of C (delivery in one year)	\$1,100
Interest rate	5%
Transaction	Cash Flow
$t = 0$	
Borrow \$1,000 for one year at 5% interest	\$ 1,000
Buy α amount of C in spot market for \$1,000	(\$1,000)
Sell a futures contract for \$1,100 for delivery	\$ 0

Total Cash Flow	\$ 0
$t=1$	
Remove C from storage	\$ 0
Deliver the good against the futures contract	\$1,100
Repay loan including interest	\$1,050

Total Cash Flow	\$ 50

Table 1.2: Cash-and-Carry Arbitrage Transactions (Overdahl and Overdahl, p. 36)

The above arbitrage opportunity can only be prevented if the futures price of commodity C of amount α is equal to or less than the spot price at $t=0$ plus the

interest rate from the loan. Hence from the example above we can derive the first rule of cost of carry: The futures price of the commodity plus the carrying charges (interest rate) to hold and deliver the spot commodity at the contract date. The rule is expressed as follows:

$$F_{0,T} \leq S_0(1 + C) \quad (1.3)$$

where,

- $F_{0,T} \rightarrow$ the futures price at $t = 0$ for delivery at time $t = T$
- $S_0 \rightarrow$ the spot price at $t = 0$
- $C \rightarrow$ the expected cost of carry, expressed in fractional terms of the spot price, used to hold the goods forward from $t=0$ to $t=T$.

The above situation does not exist in rational markets. The cash and carry arbitrage opportunity exists as mentioned earlier only because the spot price is too low relative to the futures price. Another possibility can also exist where the spot price is too high relative to the futures price. Such a situation gives rise to the reverse cash and carry arbitrage opportunity. The steps to gain from this opportunity are opposite to those of cost of carry arbitrage (Overdahl and Overdahl, p. 36).

Illustration of the reverse cash and carry arbitrage is shown in table 1.3. Under such a situation the trader sells the commodity short. Hence, he borrows the commodity from another trader and sells it in the cash market at time $t = 0$. The proceeds from the transaction are lent out for one year at an interest rate of five percent. The trader also

Table 2.1 Reverse Cash-Carry Arbitrage Transaction (Overdahl 37)	
Prices for analysis:	
Spot price of commodity C per α amount	\$1,050
Future price of C (delivery in one year)	\$1,100
Interest rate	5%
Transaction	Cash Flow
$t = 0$	
Sells α amount of commodity C short	\$ 1,050
Lend the \$ proceeds for one year at 5%	(\$1,050)
Buys a future contract for delivery in one year	\$ 0

Total Cash Flow	\$ 0
$t=1$	
Collects proceed from the loan	\$1102.5
Accepts the delivery on the future contract	\$1,100
Repayes the C from delivery to the trader	\$0

Total Cash Flow	\$ 2.5

Table 1.3: Reverse Cash-and-Carry Arbitrage Transactions (Overdahl and Overdahl, p. 37)

buys a futures contracts for the delivery of commodity C of amount in one year at time $t=0$. At time $t=1$, collects the proceeds from the loan which include the principal amount and the interest amounting to eleven hundred and two dollars and fifty cents. Then he accepts delivery on the futures contract. Returns the commodity C to the trader from which it was borrowed at time $t=0$. The trader in the arbitrage opportunity ends up with a profit of two dollars and fifty cents. It is to be noted again that the trader had no net cash flow at time $t = 0$, therefore this strategy required no investment. Considering the case mentioned above the cost of carry rule two is

derived. The rule is expressed as follows:

$$F_{0,T} \geq S_0(1 + C) \quad (1.4)$$

Hence if the above rule is met then there is no risk of reverse cash of carry arbitrage. Table 1.4 summarizes the transactions needed to carry both kinds of arbitrages (Overdahl and Overdahl, p. 38).

Table 2.2 Transactions for Arbitrage Strategies (Overdahl 38)		
<i>Market</i>	<i>Cash and Carry</i>	<i>Reverse Cash and Carry</i>
Debt	Borrow Funds	Loan out proceeds from short sale
Physical	Buy and store asset, Deliver against futures	Sell asset short, secure proceeds from short sale
Futures	Sell futures	Buy futures; accept delivery, return Physical asset to honor short sale commitment

Table 1.4: Transaction for Arbitrage Strategies (Overdahl and Overdahl, p. 38)

In order to prevent arbitrage opportunities from arising two rules of cost of carry arbitrage must hold. Both these rules can be combined to form a third rule, which suggests that the spot price of a commodity including the carrying charges must equal the futures contract price. The rule is expressed as follow:

$$F_{0,T} = S_0(1 + C) \quad (1.5)$$

The above rule is contrived on the assumptions mention earlier that the markets are perfect, no transactions cost exist for both sides of the market, no restrictions on use of proceeds from the short sales, and that the underlying asset of the contract provides no proceeds to the owner. Considering these conditions the expected cost of carry defines the relationship between the cash and futures market. This relationship as described earlier is known as the basis. In real markets the assumption made in these model are not valid in describing the basis accurately, hence adjustments to the equation 1.5 are required(Overdahl and Overdahl, p. 39).

1.3.4.2 Cost-Of-Carry Model in Imperfect Markets

The perfect markets cost of carry model assumes no transactions costs. In real markets there are transactions cost which are realized on both the futures market and the spot market. For this model we will assume that there are no transactions cost in the futures markets, hence only the transactions costs in the spot markets will be considered. These transaction costs include brokerage fees and various exchange fees. In this analysis we assume that the transactions cost are some fixed percentage, T , of the transaction amount.

Using the figures from the perfect markets table 1.5 illustrates the impact of the transactions cost. With the transactions costs the attempted arbitrage results in a loss rather than an arbitrage profit. In the table we see that eleven hundred dollars are borrowed by the trader to cover the cost of amount of C commodity including the transactions cost at $t = 0$. At time $t = 1$ after settling the futures contract the proceeds from that do not cover the cost of the initial loan and the interest. Hence

for this arbitrage to be successful we need to pay less than eleven hundred dollars towards loan repayment. Thus given the transactions cost it does not pay attempt this cash and carry arbitrage as the trader end up with a fifty five dollar deficit. Essentially this happens because we need to pay the spot price plus the transactions costs $S_0(1+T)$ to acquire the good at time $t = 0$. Carrying the good to delivery costs $S_0(1+T)(1+C)$, These costs include acquisition of the good as well as the carrying charges to delivery of the good(Overdahl and Overdahl, p. 41).

Table 2.3 Attempted Cash-and-Carry Arbitrage (Overdahl 41)

Prices for analysis:	
Spot price of commodity C per α amount	\$1,050
Future price of C (delivery in one year)	\$1,100
Interest rate	5%
Transactions Cost (T)	10%
Transaction	Cash Flow
t = 0	
Borrow \$1,100 for one year at 5% interest	\$ 1,100
Buy α amount of C in spot market for \$1,000 with \$100 in transaction costs	(\$1,100)
Sell a futures contract for \$1,100 for delivery	\$ 0

Total Cash Flow	\$ 0
t=1	
Remove C from storage	\$ 0
Deliver the good against the futures contract	\$1,100
Repay loan including interest	\$1,155

Total Cash Flow	(\$ 55)

Table 1.5: Attempted Cost of Carry Arbitrage Transaction (Overdahl and Overdahl, p. 41)

Thus in order to break even the transactions cost must yield eleven hundred and fifty five dollars. This can be formally written as:

$$F_{0,T} \leq S_0(1 + C)(1 + T) \quad (1.6)$$

If the equation inequality above holds then there is no arbitrage opportunity that is available for the trader. Extrapolating from the perfect market model, equation for the reverse cash-carry arbitrage opportunity can be derived as follows:

$$F_{0,T} \geq S_0(1 + C) \quad (1.7)$$

This equation was derived to counter act the reverse cash-carry arbitrage opportunity in perfect markets. In imperfect market we need to adjust the equation above for the transportation costs added. The new equation for the reverse cash-carry arbitrage is

$$F_{0,T} \geq S_0(1 + C)(1 + T) \quad (1.8)$$

Hence both equation 1.7 and 1.8 can be combined together to give:

$$S_0(1 + C)(1 - T) \leq F_{0,T} \leq S_0(1 + C)(1 + T) \quad (1.9)$$

The combination of the two equations provides us with an upper and lower bound for the futures prices. Any price above or below this bound set by the two equations will be open to arbitrage opportunity. Traders would be able to make a riskless profit beyond the no-arbitrage bounds. However, the price can fluctuate between the bounds without creating an arbitrage opportunity. Figure 1.7 illustrates the said relationship(Overdahl and Overdahl, p. 45).

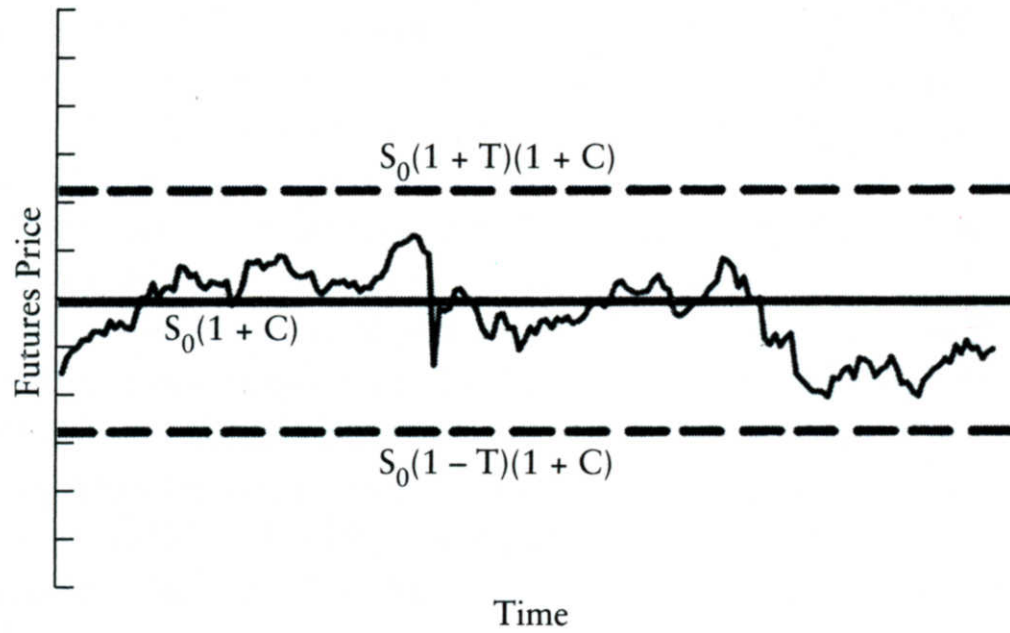


Figure 1.7: No Arbitrage Boundaries (Overdahl and Overdahl, p. 44)

So the futures price stays between the two boundaries. If the futures price crosses the boundaries, arbitrageurs will take advantage of the situation and buy spot commodity and sell the futures. Hence the price of the spot commodity will increase such that it is no longer feasible to do so.

1.4 Traders

People who trade derivative contracts in commodity exchanges are called traders. People don't have to be physically present on the floor of the commodity exchange to be called traders. Anybody who trades contracts either himself or through another person is called a trader. There are two basic kinds of traders hedgers and speculators. These two groups of people have different motives in trading in the commodity exchanges. "Hedgers" are people who seek to minimize and manage price risk, while "speculators" are people who are willing to take on risk in the hope of making a profit.

Any person who holds a position on the commodity exchange which is equal but opposite to a position held in the physical commodity is said to be hedged. Suppose this hedger is a farmer who expects to harvest 5,000 bushels of soybean in the future. He can use the futures contract to lock in a price that is feasible for him to make a profit. Like this farmer who is hedging against a price decrease is said to hold a short position on the exchange, but holds a long position in the cash market. The hedge has little concern about the future price since he or she already effectively sold this future product at a fixed price.

A non-hedged "long position", means that the farmer individual will benefit if the price of soybean rises in the future. By contrast, a "short position" means that the individual will benefit if the soybean prices drop. Therefore the individual will be better off if the price of the soybean increases in the spot or cash market thus has a long soybean in the cash position. If he or she currently expects the price to go down

he might open a short position in the futures exchange for five thousand bushels of soybean. If the price drops 10 cents per bushel in the cash market where the soybean will be sold, then the farmer in this position would incur a loss of five hundred dollars. This loss will be offset by the short position in the futures contracts. Suppose the price of one bushel of soybean was eight dollars, and now it has dropped to \$7.90. This means that the farmer can buy back his position at \$7.90 per bushel and close the contract. Therefore the actual beans can be delivered by the farmer to the market at the \$7.90 per bushel, and he will cover his potential loss by the difference paid by the person who bought the original futures contract from the farmer \$8.00 per bushel(Gould, p. 71).

Now suppose that the hedger is not a farmer, rather a person who owns a soybean processing plant. This person has a short position in the cash soybean market. Therefore, in order to hedge against a price increase in the cash market he will buy a long position in the futures exchange. This is often termed as “buying hedge”, as opposed to the farmer’s “selling hedge”. In this case, the business owner will receive the difference between the price of the futures contract and the spot price to buy his or her commodity in the spot market. For example, if the price of a soybean bushel increases from \$8.00 per bushel to \$8.20 per bushel, then the business owner will make a profit of 20 which he can use to pay the difference in the spot price when he buys the soybeans for \$8.20 per bushel rather than\$8.00 per bushel.

Speculators in the commodity markets trade for the quick rewards. They analyze the market to best determine the direction of the future price of a contract. They may also engage in the study of external events that affect the price or they may

use historical price movements to predict the current price. Often times, speculators specialize in some particular commodity. These traders make a profit on their projections of the price movement in the future. A speculator who anticipates an increase in price would want to take advantage by buying a future contract. If the trader expects the price to decrease he will sell the contract at the higher price, and buy it later at a lower price. Speculators have no physical commodity in hand as they trade contracts in the futures exchange market. They always try to be in the long position when the prices rise and in short position when they fall. Their positions are always cash settled(Gould, p. 75).

In the past, traders came from a select social and economic category in the society, mostly concerned with agriculture products. Today the traders are very diverse. They include large corporations, industrial firms, and investment institutions which have joined in the trading arena. The firms trade in the market to get a lower price for their raw material. Increase in the variety of contracts has provided traders with more possibilities for hedging and profit making through speculation. Most of the individuals and firms trade through brokerage houses who have memberships in the commodity exchanges. Representatives of these brokerage firms trade on the exchange floor or pit on the behalf of their clients. Some traders get memberships to the commodity exchanges for themselves and trade in the pit for personal gains. These individuals are known as scalpers(Gould, p. 76).

Scalpers are a type of speculators who help the market reach it's clearing positions. They make their profits on very small market discrepancies. These discrepancies can be as small as a quarter of a cent. In case of these discrepancies the price of futures

contract has fallen below, or risen above a fair and reasonable estimate of the cash price. In an active pit ten or twenty seconds may mean two or three fluctuations in the price. Therefore, they are also responsible for the liquidity of the contracts in circulation. They can afford to take advantage of these small price movements because they pay no commissions, unlike other speculators, who must factor commission cost into their profit projection(Gould, p. 75).

1.5 Soybean Commodity Markets

To study the efficiency of the Commodity Markets, soybean complex was chosen due to its importance in the feed production for livestock, use in chemicals, and ingredient in the vegetable oil. Soybeans need to be processed into oil and meal before they can be used for commercial purposes. The price of the soybean complex is therefore determined by the combined price of soybean oil and meal including the processing and handling costs.

Approximately 150 varieties of soybeans are grown in the United States (US). Most of these varieties are created through genetical modification. Soybeans grow best in fertile and sandy soil suitable for corn and cotton. Planting season starts in late May or June after the corn planting season. The soybean plant is photoperiodic, meaning that ripening and flowering of the soybean plant are relative to the length of daylight it receives. Therefore, soybeans are genetically modified so that the plant can effectively grow in varying lengths of day light(CBOT, p. 157).

In the past, late plantings of soybeans resulted in smaller harvest yields. Today

genetically modified varieties of soybean can grow in latitudes from Minnesota to Mississippi. These genetically modified soybeans also enable farmers to harvest different batches of soybeans within a few days of each other regardless of the time they were planted. Thus, genetical modification of the soybean plant has helped to increase the amount of soybean harvested per acreage throughout the US.

Today the United States is one of the leading producers of soybeans. The amount of soybeans produced in the US is more than the amount that can be consumed by domestic markets, therefore much of the soybeans that are grown in the US are exported. The dollar amount of the exported crop usually amounts to billions of dollars. The main importers of soybean include the European Union (EU) and The Peoples Republic of China. The importing quotas and grade restriction in these countries play an important role in price development of soybeans in the US and consequently its products meal and oil. Since US production of soybean heavily depends of genetically modified soybeans to increase yield, protein, and oil content thus before soybean can be exported, they need to be approved for import by the EU and the Chinese trade commissions(CBOT, p. 158).

Hence, high yield does not necessarily indicate high exports, though a high yield impacts the domestic prices of soybean. Yields of other countries such as Brazil also affect the world market price of soybean, but since the US is the largest exporter therefore US exports have a greater impact on the international price of soybean. The US produces only one soybean crop per year compared to three in South America. Hence, a larger yield in other counties can lower US domestic prices as much of the

access grain in the US cannot be exported at competitive prices. Thus the amount of soybean exported is an important price indicator for the soybean commodity markets.

Soybeans are most popular in the areas where corn and cotton are grown. Corn and cotton are nitrogen intensive crops, therefore they consume the nitrogen present in the soil. Corn or cotton crops cannot be grown for consecutive years as the their yield slowly degrades with the depletion of nitrogen from the soil. Thus, soybeans are used as an interim crop after cultivating two years of corn or cotton, nitrogen intensive crops, because soybeans, being a legume, return nitrogen back into the soil. Soybeans are also a substitute for corn in the oil market, and the soybean meal is used to feed livestock instead of the corn stalk. Since corn and soybeans are substitutes of each other therefore the price of one can be used to predict the price of the other.

The stability in prices is important to both the farmers and the processors of a given commodity. As development in agriculture has enabled farmers to produce a higher yield per acreage, which causes major price fluctuations in crop prices from one year to the next. Thus the US government has price support programs for the farmers to ensure price stability from one year to the next. The government price support programs fall into two main categories. First the government can directly purchase the produce of a farmer or provide him/her with a subsidy payment making up the difference between the market and parity price. The parity price is the adjusted price of a commodity in terms of the price on non-agricultural products as they were in some given base year. Lets suppose that the price of ten bushels of soybeans bought ten gallons of crude oil in 1950. Therefore accordingly the parity price today should be set where 50 bushels of soybeans should buy gallons of crude oil. Unfortunately

from 1950 till today the efficiency in farming has increased more rapidly than other industries as afore mentioned advances in genetics thus it hard for the government to maintain the parity price(Gould, p. 183).

The second price support program which is provided by the government is the loan program through the United States Department of Agriculture (USDA). The loan program is the most effective government policy today price stabilization for the farms which has the most profound effect on the cash and futures markets. Before the planting season begins, the USDA announces its “loan rates” for each of the crops. For the farmers the loan rate is the lowest price they will get for their produce, thus they can plan their planting accordingly. Suppose the loan rate represents \$2.50 per bushel. This ensure the farmer that the harvest time spot price for that particular crop will be at least \$2.50 per bushel. The farmer has the opportunity to borrow money for machinery rental, labor, and for other inputs against his crop for the upcoming planting season. If he is unable to pay the loan back by selling his crop in the cash market, then he will just give up his crop to the USDA, and keep the loan money. Otherwise he can sell his crop in the cash market, keep the profits, and repay the loan to the government. Therefore the loan rate becomes an effective price floor for a particular crop for that given year. The spot and futures market incorporate this price floor into their valuation of the crop prices (Gould, p.185).

The soybeans are useful for commercial purposes only after they have been crushed into meal and oil. Crushing refers to the processing of soybean where the products of the processing procedure are oil and meal. The soybean meal is crushed into either a high protein feed for livestock, or further processed into soy grits, flour, or isolated

protein. The crude oil created by the chemical extraction process can be processed into edible mayonnaise, salad oil, cooking oil, and margarine. The processing of soybeans into these different products depends on the difference between the prices of soybeans and its crushed byproducts meal and oil. This price differential is known as the crushing spread. The profitability of soybean processors is measured through this crushing spread. This spread measures the cost of soybeans in relation to the sales return on the processed soybean oil and meal. So the price of the soybean meal and soybean oil cannot be lower than the price of the soybeans. The price of meal and oil are generally higher than the soybeans by at least the amount it takes to process soybeans into meal and oil. Since the soybeans cannot be used as beans therefore the price of the beans are also dependent on the supply and demand of the meal and oil. If meal and oil prices rise due to a high demand then the prices for soybeans will also rise. Therefore the prices of soybeans and its processed products meal and oil move in relation to each other. If the cost of the soybeans exceeds the returns on the sale of the processed soybean oil and meal, then it is no longer profitable for the processor, thus resulting in a slow down of production. Therefore the prices of the soybeans are important in determining the level of production of soy oil and meal(CBOT, p. 158).

There is no set rule for a profitable crushing spread for the processor. Generally it is agreed that a margin of 15 cents per bushel is the lower limit beyond which there is no profit to be made. This would mean that the returns from the sale of both the oil and meal would generate a profit of 15 cents per bushel when the price of the input, soybean is taken away. For example:

Assume that the price of soybean oil is \$15.00/hundred pounds or 15

cents per pound. The relative yield of soybean oil per 60 pound bushel of soybean is 11 pounds. Therefore the price of soybean oil per bushel would be $15 \times 11 = \$1.65$. Also assume that the price of meal is 10cents/pound. Generally, 47 pounds of meal is produced from a 60 pound bushel of soybeans. Therefore, the price of soybean meal per bushel would be $47 \times 0.1 = \$4.7$. If the price of the soybeans was \$6.10 per bushel then the crushing spread would be $\$1.65 + \$4.7 - \$6.10 = \$0.25/\text{bushel}$. This is a profitable crushing spread for the processors since the lower limit for profitability is 15 cents(CBOT, p. 160).

There is a seasonal trend in the spot soybean prices and consequently in the crushing spread. During the early months of the harvesting season the prices of the soybean are at their lowest levels, and the crushing spreads reflect a favorable profit margin. The demand for soybean meal is high during the same period since winter is approaching and due to cold weather and lack of grazing pastures the demand for feeding requirements for livestock increases. The combination of excess supply of soybeans, and high demand for soybean meal help create a profitable environment for processors to make good profits. This spread tends to shorten during the later part of the year as demand for soy meal decreases, leading to lower prices. The supply of soybeans also diminishes during the year and therefore the prices of soybeans increase. A higher than expected yield of the soybean crop will produce profitable crushing spread, while a low yield of the crop will produce unprofitable crushing spreads. Low crushing spread does not mean that companies that process soybeans will shutdown. Instead they decrease their output to reduce supply, thus raising the

price of soybean meal and oil to sustain through these low periods. Soybeans are the easily storable form of the commodity. Once soybeans have been processed more care is required in their storage and therefore higher costs are incurred(CBOT, p. 159).

1.6 Commodity Market Efficiency

The price of the crops during the year remains a concern for the producers of crops, farmers and the owners of processing plants and other industries related to agriculture as presented in the earlier section. Price fluctuations can cause people unexpected financial losses. Thus they hedge in futures markets in order to set in a price at the time they need to sell or buy a commodity. The price of the futures contract is a prediction of the spot price of the commodity at the expiration date of the contract. The concern that hedgers and speculators have is whether the futures contracts are correctly priced such that they reflect all available information about the future spot price of the commodity when the contract expires.

“Efficiency, therefore, refers to the two aspects of a price adjustment to new information, the speed and quality (direction and magnitude) of the adjustment”(Keane, p. 9). The efficiency that this study is concerned with is informational efficiency of the market. Houthakker and Williamson (1996) defines a market as informationally efficient if at any time (assuming the market is open for trading) all information available at that time is fully reflected in current prices. This efficiency must not be confused with operational efficiency. A market is considered operationally efficient if the transaction costs for trading is the lowest possible. Houthakker and Williamson

(1996) states that operational efficiency of a market is the pre-requisite for informational efficiency. Since traders cannot act on newly available information to effect the price of an asset if the trading costs are high. Determining if the market is operationally efficient is out of the scope of this study, therefore is assumed that commodity markets are operationally efficient.

The concept of informational efficiency is explained through rational expectations theory. The application of the rational expectations theory in financial markets such as the commodity market can be analyzed through the efficient market theory. Both theories combined conclude that if the markets are informationally efficient then any trader in the market cannot make abnormal profits consistently. The level of efficiency of the market will then be presented at the end.

1.6.1 Theory of Rational Expectations

Theory of rational expectations is a broader concept of which the efficient market theory is a subset. The theory of rational expectations also known as the rational expectations hypothesis (REH) was developed by John Muth. In broader terms the theory states that, “in a competitive world, economic agents will exploit all available information to take advantage of any perceived profit opportunities”(Keane, p.13). The theory thus implies that even though economic agents cannot accurately predict the state of the world in the future, they do not make systematic mistakes in their forecasts. Therefore, the drive for profit making by economic agents will tend to remove any obvious opportunities for abnormal profit(Keane, p.13).

Mishkin (1997) states that according to REH the “expectations [about the future] will be identical to optimal forecasts (the best guess of the future) using all available information” (p. 690). This statement can be explained through an example of soybean spot price forecasting. Suppose that the price of fuel has a direct effect on the prices of soybeans through an increase in transportation costs. Hence, if the prices of fuel stays constant then the price spot of in Chicago will move in an upper of lower bounds of \$1.00. Therefore, suppose that the price of soybeans in the beginning of the a given week is \$3.00 per bushel then the highest it can fluctuate within that week is \$4.00 per bushel, and the lowest it can reach \$2.00 per bushel. Now also suppose that if the price of fuel increases by 10 cents per gallon then there is a general increase in the prices of soybean of about \$1.00 per bushel in Chicago due to higher transportation costs. So if yesterday’s price of soybeans was \$3.00, and if the price of fuel has risen today by 10 cents per gallon, then the best guess or the *optimal forecast* of the price of soybeans will be \$4.00 because on average the price of soybean remains at \$3.00 and that a 10 cent increase in fuel would cause a general increase in soybean prices of a \$1.00. So the best guess or expectation by REH using all available information, the average price of soybeans in a given week, and the rise in prices of fuel, would be \$4.00.

In the example above our best guess was \$4.00. After today’s trading it could happen that the price of soybean might be \$5.00 or \$3.00 considering the dollar upper and lower movement bounds. Nonetheless, the best possible expectation of the price which correct on average would still be \$4.00. This illustrates an important point about the rational expectations, “ that even though the rational expectation equals

the optimal forecast using all available information, a prediction based on it may not always be perfectly accurate” (Mishkin, p. 690).

Consider a situation where relevant information might not be available or is simply ignored. For instance that there is a fungus that has destroyed much of the stored soybean crop leading to a general price increase of about \$2.00 per bushel. In light of this new piece of information the optimal forecast should be \$6.00, but since the information was not incorporated the expectations of today’s price therefore the expectation is not rational. Therefore Mishkin (1997) points out that there are two reasons why expectations may fail to be rational.

- 1 - People might be aware of all available information but find it takes too much effort to make their expectation the best guess possible
- 2 - People might be unaware of some available relevant information, so their best guess of the future will not be accurate

Mishkin (1997) further states that even though an additional factor is important in making the best guess, but information about it is not available, then still an expectation that does not take into account the information, can be rational because an expectation is rational if it uses all *available* information.

Formally the theory of rational expectations states that, if χ stands for the variable that is being forecasted, χ^e for the expectations of this variable, and χ^{of} for the optimal forecast of χ using all available information, then

$$\chi^e = \chi^{of} \tag{1.10}$$

That means that the expectations of the variable χ will be equal to the optimal forecast χ^{of} using all available information(Mishkin, 691).

The question that needs to be answered is why do people want their expectations of the future to be optimal forecasts. In financial markets such as the commodity markets “people with better forecasts of the future are able to make more money than people who don’t”(Mishkin, 691). This brings about two implications of the rational expectations theory.

- 1 - “If there is a change in the way a variable moves, the way in which expectations of this variable are formed will change as well”(Mishkin, 691). This means that due to the fungus epidemic the price variability in a week is only found to be increasing. Therefore, the expectations of economic agents will change from the dollar variation below and above \$6.00 to only a dollar above \$6.00.
- 2 - “The forecast errors of expectations will on average be equal to zero and cannot be predicted ahead of time”(Mishkin, 691). The error in forecasting variable χ defined above is equal to $\chi - \chi^e$, which is defined as the difference between the expected value of the variable and the actual value of the variable. Now suppose that the average expected forecasts of the actual value of the variable are off by one dollar, then the economic agents will adjust their expected forecasts by one dollar such that the average expected forecast error is equal to zero. Since the average expected forecast error is equal to zero, thus it cannot be predicted a head of time.

The above two implications of the theory of rational expectations explains how even

investors without private information can infer that information from the movement of prices. Suppose that privately informed economic agents receives positive information about an asset or in the case of commodity markets the amount of soybean reserves, stocks. The increase in demand of these economic agents will cause the spot price of soybeans to increase. The uninformed economic agents will observe this increase in price and infer from it that privately informed agents received positive information. Thus, the prices will adjust as if all the agents in the market had the same information(Dyckman and Morse, p. 18).

The theory of rational expectations implies that since all the agents in the market strive to make their expectations be equal to the optimal forecast using all available information, therefore no agent can make abnormal profit. Even if they are privy to privately information still they will not be able to make abnormal profits consistently. This theory is not testable unless it has some empirical content. The expectations of individuals in the market cannot be gauged and it cannot be related to the optimal forecast. Since both these are unobservable phenomena. The application of the theory of rational expectations can be found in the efficient market theory.

1.6.2 Efficient Markets Theory

The efficient markets theory is essentially the application of rational expectations theory in financial markets. The efficient market theory is based on the assumption that prices of assets in the market fully reflect all available information. Similar to the rational expectations theory, the efficient market theory states that no economic agents can make abnormal returns in the market. The efficient market theory also

takes into account the intrinsic value of an asset. If the price of the asset deviates from this intrinsic value then the asset is consider over or under priced. Dyckman and Morse (1986) states that having a return model is necessary to establish the link between expected returns and actual returns, and without this model, the efficient market hypothesis cannot be tested. The model defined by Mishkin (1997) is presented below which will be used as a bases for testing the efficiency of the commodity markets.

The equation below is formed through the evaluation of expected returns and actual returns realized for a given asset. The return on holding any given asset should be equal to the change in its price, capital gains:

$$RET = \frac{P_{t+1} - P_t + C}{P_t} \quad (1.11)$$

where,

RET = rate of return on the security held from time t to $t+1$

P_{t+1} = price of the security at time $t+1$, the end of the holding period

P_t = price of the security at time t , the beginning of the holding period

C = cash payments (coupon or dividend payments) made in the period t to $t+1$

Incase of commodity futures there are no cash payments involved. Thus, for commodity markets $C = 0$.

The expectations of the returns RET at the beginning of the hold period are denoted as RET^e . The current price of the security is known at time t and $C = 0$. Thus the only thing that is for uncertain at time t is the price P_{t+1} . The expected

return at $t+1$ can be written as

$$RET^e = \frac{P_{t+1}^e - P_t}{P_t} \quad (1.12)$$

Mishkin (1997) states that according to efficient market hypothesis the expectations of the future price of an asset should also be equal to the optimal forecast using all available information. Therefore it stipulates that the expectations of future price of an asset should therefore be rational; that is

$$P_{t+1}^e = P_{t+1}^{of} \quad (1.13)$$

The above statement implies that if the expectations of the future price are rational then the expectations of the returns must also be rational since the expected price in the future determines the expected return on the asset, so

$$RET^e = RET^{of} \quad (1.14)$$

Neither the RET^e nor P_{t+1}^e can be observed thus another method of evaluating RET^e must be used. Mishkin (1997) shows that the expected return on a asset is equal to the equilibrium return on the asset, through an analysis of supply and demand of the asset. That is represented as

$$RET^e = RET^* \quad (1.15)$$

Since the rational expected return on the asset is equal to the optimal forecast of the return, therefore the equation above can be re-written as

$$RET^{of} = RET^* \quad (1.16)$$

The equation above tells us that the “current prices in a financial market will be set so that the optimal forecast of a security’s return using all available information equals the security’s equilibrium return”(Mishkin, p. 694).

The equations 1.14,1.15,1.16 tell us that a disequilibrium in these conditions would be an opportunity for an economic agent in the market to make profit. Lets suppose that the return on a futures contract for soybeans in the month of June is five percent, and the price of the contract to P_t is less that the optimal forecast, P_{t+1}^{of} , such that the RET^{of} is equal to 10%. This is an opportunity for an economic agent in the market to make a profit, since the $RET^{of} > RET^*$. Thus the economic agent will buy more soybean contract thereby raising the demand of the contract. The increase in demand of the contracts will increase the current price, P_t of the contract. The increase in the current price will decrease the optimal forecast return, RET^{of} . The agent will buy enough contract where the optimal forecast return is equal to the equilibrium forecast return as shown in equation 1.16. The reverse would be true if the optimal forecast return was less than the equilibrium forecast return. In that case the economic agent in the commodity market will sell soybean commodity contracts until the optimal forecast return became equal to the equilibrium return(Mishkin, p. 695).

The situation above can be simplified as follows:

$$RET^{of} > RET^e \rightarrow P_t \uparrow \rightarrow RET^{of} \downarrow$$

$$RET^{of} < RET^e \rightarrow P_t \downarrow \rightarrow RET^{of} \uparrow$$

The above will two scenarios will take place until $RET^{of} = RET^*$.

According to Mishkin (1997), the above condition of $RET^{of} = RET^*$ allows us to simplify the efficient market condition as follows:

- In an efficient market, all unexploited profit opportunities will be eliminated
- Not everyone in a financial market must be well informed about a security or have rational expectation for its price to be driven to the point at which the efficient markets condition holds

This implies that not all of the market participant need to exploit the profit making opportunities. Only a few individuals in the market can make the market efficient because of the incentive of making profit. Since the market will be efficient after the profit opportunity is eliminated therefore there is no chance for an economic agent to make these abnormal profits consistently. Hence, an efficient commodities market will imply that future contract prices will be an unbiased predictor of the future spot prices, using all available information.

1.6.3 Levels of Market Efficiency

The three levels of market efficient where presented by Eugene Fama (1970). The work of Fama on the theory of market efficiency is known as the efficient market hypothesis (EMH). The EMH has three progressively stronger forms of market efficiency: weak form, semi-strong form, and strong form. The diagram below illustrates the relationship:

The weak form of the efficient markets hypothesis suggests that past prices, volume, and other market statistics provide no information that can be used to predict

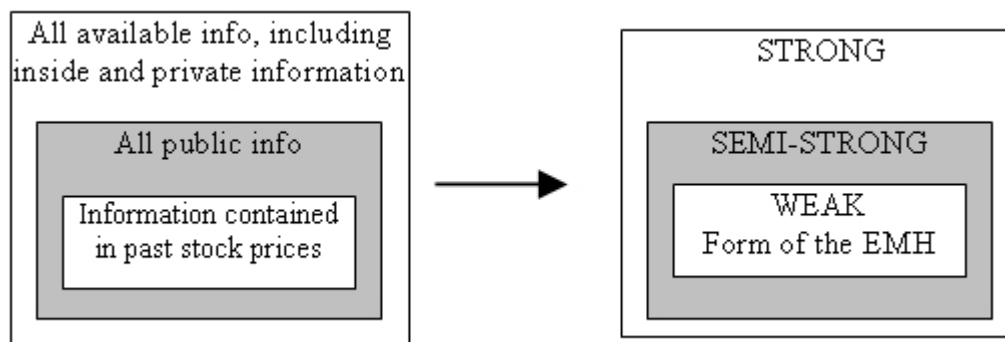


Figure 1.8: Levels of Market Efficiency in Relation to Information

future prices. This would suggest that in commodity markets, the spot price fluctuations are random, and can not be used to predict the future spot prices. Therefore past spot prices cannot be used to predict future spot prices, also the past prices of futures contract cannot be used to predict future contract prices if the weak form of the EMH holds. Thus price changes are random because these changes are based on random information which is available only overtime. Therefore charting techniques which use historical prices as an indicator of the future price movement will not provide any new information that can be used to “beat the market”. “Beating the market” means that an individual gains extraordinary profits from buying or selling a particular asset. There is also no gain to be made in the technical analysis of information such as expected harvest yield, transactions costs, and storages costs because these are already factored into the current prices. People who expect to gain

from technical analysis believe that the market is mostly psychological and thus past behavior predicts future behavior. If the weak form of the EMH does not hold, then the market would show a herd mentality. If some individuals buy a futures contract at a certain price, then the rest of the traders would follow (Keane, p. 10).

The semi strong form of the efficient markets hypothesis incorporates the assumptions made in the weak form of the EMH, that current prices reflect all publicly available information, including historical prices, macroeconomic data, and expectations about the future. “A market is efficient in the semi-strong sense if asset prices respond instantaneously and without bias to newly published information” (Keane, p. 10). If this form of the hypothesis holds, then there is no point incorporating new news in the evaluation of the asset price because the current price already reflects this relationship.

The strong form of the efficient market hypothesis implies that the prices fully reflect all information whether the information is publicly available or not. Even the knowledge of material, non-public information cannot be used to beat the market. In case of financial markets, the strong form of EMH includes the evaluation of insider information into the valuation of assets.

The challenge in efficient markets is how quickly the market prices reflect new information. Figure below illustrates the reaction of the traders to new information.

Markets only efficient in the weak form of EMH will have an under reaction to the new news that has been made public. The price of the asset will rise slowly from P_{PRE} to P_{POST} . In case of the semi-strong market there will be a gradual rise from

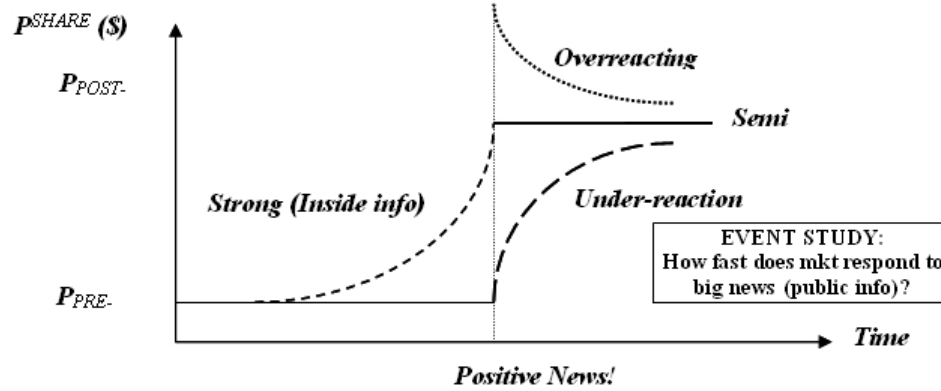


Figure 1.9: Types of Reactions to New Information

P_{PRE} to P_{POST} . If the market is efficient as per the conditions of the strong EMH, then the prices will instantaneously jump from P_{PRE} to P_{POST} .

This study will challenge the weak hypothesis as the study is trying to use past data to predict the spot prices in the future. Based on the prediction of the spot prices a trading rule can be generated. The performance of the trading rule in terms of making money will enable us to make the judgment whether the abnormal profit can be made consistently. If they can then the weak form of the efficient market hypothesis will not hold for Market Efficiency. Therefore we can suggest that according to data the futures markets are inefficient in making optimal forecasts of spot prices in the future.

CHAPTER 2

Literature Review

The focus of this chapter is to review scholarly literature on testing commodity market efficiency. Two articles were chosen based on their empirical and theoretical relevance to the models being used to test market efficiency as per the efficient market hypothesis. Study performed by Rausser and Carter (1983) shows that the soybean and soybean meal markets are not semi-strong efficient as presented by Fama (1970). The model he uses to test for the inefficiency uses indicators of spot prices that can also be used by this study. The other study is performed by Kenyon et al. (1993). He tests the efficiency of the futures market contracts of May in predicting the spot price at harvesting time.

2.1 Article 1: Futures Market Efficiency in the Soybean Complex

The article by Rausser and Carter (1983), “Futures Market Efficiency in the Soybean Complex” claims that past research has been without a sound foundation since their empirical work has concentrated on a search for a random walk or generally the Martingale properties of future contract prices. The articles states that previous studies have concluded that periodic failure of the Martingale property for futures market is not evidence of the inefficiency in the futures market. Carter stipulates that failure of the Martingale property does not have a direct relation to the efficiency of the futures markets. Therefore, Carter examines the efficiency of the futures market using the semi-strong efficient market hypothesis by Fama (1970). He investigates the ability of the futures market to forecast the spot price at harvesting time. The study examines the soybean, soybean oil and soybean meal separately. It concludes that the soybean and soybean meal are not efficient under the semi-strong hypothesis, but the soybean oil commodity market is.

Review of the past literature by the authors indicates that there is a debate on the efficiency of the futures markets. Some studies conclude that the futures markets are efficient while other conclude they are not, and some are inconclusive. Thus, if the futures market is infact efficient then these markets are a relevant price signal to the producers of the commodity. Carter also mentions that the past literature assumes that the risk in the basis does not exists. Meaning that the difference between the spot price and the futures price is non-existent. This As mentioned earlier, the past

literature looks for the inefficiency of the market in terms of the Martingale process or specifically the random walk theory. The study performed by Carter, provides a more powerful test of commodity market efficiency than that of earlier studies which perform weak hypothesis test of Fama in the search for a random walk. The study by Carter on the other hand uses the semi-strong hypothesis provided by Fama to test the efficiency of the commodity futures market.

The purpose of the study performed by Rausser and Carter (1983) is to find whether the futures market is efficient as per the semi-strong hypothesis of market efficiency proposed by Fama(1970). This study argues that the efficient market hypothesis (EMH) framework is a more powerful test of efficiency of the markets than empirical test performed in earlier studies which test for the weak form of the EMH. The weak form test were generally geared towards testing for random walk or Martingale properties in the futures markets. Rausser and Carter (1983), stipulate that studies that the, “periodic failure of the Martingale property to hold is not evidence of market inefficiency” and “the link between a Martingale process and efficiency in futures market may be broken”(469). The case against the search for Martingale properties is found in the feedback from historical prices of the futures which affect the current prices of the futures. Therefore, the authors of this article conclude that there isn’t a direct implication of the Martingale property on the efficiency of the commodity markets.

To test for the efficiency of the commodity markets the authors propose two conditions. Relative accuracy and the incorporation of cost/benefits adjusted for risk.

These two conditions must be accounted for in the model created to test for the efficiency of the commodity markets. Relative accuracy exists if the model created by the study is more accurate in predicting the spot prices of the commodities than the futures market. “This condition is only necessary; inefficiency implies that a model does exist whose forecasts are more accurate than the future market forecasts.(Rausser and Carter, p. 469)” Further more, sufficiency is to be obtained by weighting in the marginal costs and benefits of using such a model and the risk involved in its use. These two conditions “relative accuracy and cost/benefits, are necessary and sufficient for the inefficiency property of the commodity futures markets” (Rausser and Carter, p. 469).

The relative accuracy condition of the soybeans, soybean oil, and soybean meal is tested through structurally based ARIMA models. These models are used to investigate time series data that is dependent on its past values. The study has “structural monthly model of the US soybean system and subsequently derive and estimate the associated multivariate transfer functions and the univariate ARIMA processes implied by the structural specifications”(470). These models predicts the price of the soybeans using the price of the soy oil and meal, stored soybean price, and the price of the exported soybeans. The price of the soybeans is derived from the soybean meal and soy oil which are the two derivatives of the soybeans. Soybeans themselves have no value because they cannot be used as beans. Thus, “the value of the soybeans is determined directly from the soybean meal and the soybean oil prices and also marginally by crushing and handling costs” (Rausser and Carter, p. 471). There are

a lot of substitutes for soybean oil therefore the price of the soybean oil are determined in relation to its substitutes, other vegetable oils. On the other hand, the price of the soybean meal is related to its supply and demand, and depends on the price of corn.

The econometric multivariate and univariate models used to predict the soybean prices are as follows: *Price of Soybeans:*

$$PS_t = \theta_{11} + \alpha_{12}PO_t + \alpha_{13}PM_t + \alpha_{14}(L)SS_t + \beta_{11}(L)ES_t + u_{1t} \quad (2.1)$$

Price of Soybean Oil:

$$PO_t = \theta_{21} + \alpha_{25}(L)SO_t + \alpha_{27}CS_t + \beta_{22}(L)OI_t + \beta_{23}(L)EO_t + u_{2t} \quad (2.2)$$

Price of Soybean Meal:

$$PM_t = \theta_{31} + \alpha_{36}(L)SM_t + \alpha_{37}CS_t + \beta_{34}(L)EM_t + \beta_{35}(L)PC_t + u_{3t} \quad (2.3)$$

Stocks of Soybean:

$$SS_t = \theta_{41} + \alpha_{41}PS_t + \alpha_{44}(L)SS_t + u_{4t} \quad (2.4)$$

Stocks of Soybean Oil:

$$SO_t = \theta_{51} + \alpha_{52}PO_t + \alpha_{55}(L)SO_t + \alpha_{57}CS_t + u_{5t} \quad (2.5)$$

Stocks of Soybean Meal:

$$SM_t = \theta_{61} + \alpha_{63}(L)PM_t + \alpha_{66}(L)SM_t + \alpha_{67}CS_t + u_{6t} \quad (2.6)$$

Soybean Crushing:

$$CS_t = \theta_{71} + \alpha_{71}PS_t + \alpha_{72}PO_t + \alpha_{73}PM_t + \alpha_{74}(L)SS_t + \alpha_{76}(L)SM_t + \alpha_{77}(L)CS_t + \beta_{72}(L)OI_t + u_{7t} \quad (2.7)$$

where L is the Lag operator which is defined as $L^n P_t = P_{t-n}$, and where

Endogenous Variables

PS_t = price of soybeans

PO_t = price of soybean oil

PM_t = price of soybean meal

SS_t = stocks of soybeans

SO_t = stock of soybean oil

SM_t = stock of soybean meal

CS_t = soybean crushing

Exogenous Variables

ES_t = soybean exports

OI_t = crude vegetable oil price index

EO_t = soybean oil exports

EM_t = soybean meal exports

PC_t = price of corn

LI_t = livestock price index

The data used to test the models was taken from two different sources. The data on the monthly average soybean cash prices (PS_t), soybean oil prices (PO_t), US soybean exports (ES_t) where all obtained from the Chicago Board of Trade. The crude vegetable oil prices also in the average monthly form where obtained from

the Bureau of Labor. The model was built on 132 observations, and its accuracy was tested on the last 36 observations which were saved to test the accuracy of the forecasts of the model.

The model developed on the 132 observations was tested for significance of the coefficients through t-tests, and chi-square statistics. The t-test were all significant according to the authors although the level of significance chosen was not mentioned. The critical value of the Chi-square was 32.7, and the value of the χ^2 for the soybean prices was equal to 22.04. Thus, Chi-square statistic was significant for 21 degrees of freedom for the soybean prices. The χ^2 for soybeans univariate equation was 17.28 and was significant for 21 degrees of freedom. The χ^2 for soybean oil and soybean meal was 27.01 and 13.74 respectively and they were significant for 24 degrees of freedom as well.

The forecasting evaluation of the models derived in the study was performed through the Theil's inequality coefficient. This coefficient is derived from the mean-square prediction error (MSE). The equation for MSE is as follows:

$$MSE = \frac{1}{n} \sum_{t=1}^n (P_t - A_t)^2 \quad (2.8)$$

where P_t is the predicted value, and A_t is the actual value. The Theil's inequality theorem as defined earlier evaluates the spread of the MSE error of the actual values.

$$U^2 = \frac{\frac{1}{n} \sum_{t=1}^n (P_t - A_t)^2}{\frac{1}{n} \sum_{t=1}^n (A_t)^2} \quad (2.9)$$

“The coefficient ranges from zero to infinity and is equal to one for a random walk forecast”(Rausser and Carter, p. 473). The forecasting accuracy increases as U

decreases. The most interesting part of using the Theil's inequality coefficient is that it allows you to break down your forecast error into three proportions: bias proportion, variance proportion, and covariance proportion. This allows you to see the source of your forecast error. These three ratios are given as follows:

$$1 = U^m + U^s + U^c \quad (2.10)$$

where, U^m - measures unequal tendencies between the actual (A_t) and predicted (P_t).
 U^s - measures unequal variation between the actual and forecasted price changes.
 U^c - measures imperfect covariation between the pairs of predicted covariation between the pairs of predicted and actual price changes.

Therefore, for a good forecasting model, it is ideal to have low U^m and U^s , but U^c cannot be avoided as it measure the random error present in forecasts.

The results from the general forecasting analysis using the Theil's inequality theorem are given in the Table 2.1 below (Rausser and Carter, p. 497). This table reports forecast quality statistics for the overall univariate equations of soybean, soybean oil, and soybean meal price forecasts as well as the futures markets forecasts.

The evaluation of the results in Table 2.1 shows that the predictability of the soybean futures market becomes worse as the time to maturity decreases. The same is noticed for the univariate models for the soybeans, and the soybean meal. The opposite is true for the soybean oil univariate model and the futures. The predicting capability of the model and the futures increases as the expiration date draws closer. The authors attribute "the deterioration of forecasting accuracy for soybeans and soybean meal ... to the peculiar feature of the forecast horizon" (Rausser and Carter,

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TABLE 3.—ACCURACY OF SOYBEAN COMPLEX PRICE FORECASTS
FUTURES MARKET VS. UNIVARIATE ARIMA MODELS
1977-1980

Model Price	Forecast Horizon (number of months away)							
	3	4	5	6	7	8	9	10
SOYBEAN								
Futures Market								
MSE	1.67	1.80	1.60	1.29	1.20	1.15	1.01	0.88
U^m	0.02	0.02	0.01	0.02	0.01	0.00	0.00	0.00
U^s	0.77	0.67	0.60	0.57	0.48	0.58	0.46	0.54
U^c	0.21	0.31	0.39	0.41	0.51	0.42	0.54	0.46
U	0.89	0.87	0.81	0.80	0.74	0.74	0.70	0.70
Univariate Model								
MSE	0.91	0.86	0.86	0.63	0.63	0.64	0.60	0.50
U^m	0.00	0.01	0.01	0.05	0.05	0.05	0.03	0.01
U^s	0.18	0.28	0.41	0.43	0.46	0.51	0.39	0.30
U^c	0.82	0.71	0.58	0.52	0.49	0.44	0.58	0.69
U	0.66	0.60	0.59	0.56	0.54	0.55	0.54	0.53
Sample size (n)	27	26	26	25	25	24	23	22
SOYBEAN OIL								
Futures Market								
MSE	17.38	20.94	22.06	17.38	19.17	19.69	20.71	19.87
U^m	0.02	0.01	0.04	0.06	0.06	0.10	0.14	0.18
U^s	0.52	0.55	0.50	0.43	0.42	0.37	0.20	0.30
U^c	0.46	0.44	0.46	0.51	0.52	0.53	0.66	0.52
U	0.96	0.94	0.96	0.90	0.97	0.96	1.11	1.09
Univariate Model								
MSE	24.11	24.24	24.24	21.69	21.75	22.38	23.20	23.81
U^m	0.61	0.60	0.60	0.60	0.60	0.60	0.63	0.69
U^s	0.05	0.12	0.13	0.11	0.08	0.07	0.05	0.03
U^c	0.34	0.28	0.27	0.29	0.32	0.33	0.32	0.28
U	1.13	1.01	1.00	1.01	1.03	1.02	1.18	1.19
Sample size (n)	31	30	30	29	29	28	27	26
SOYBEAN MEAL								
Futures Market								
MSE	1,559	1,958	1,704	1,512	1,377	1,262	1,046	1,015
U^m	0.01	0.00	0.02	0.01	0.01	0.04	0.08	0.17
U^s	0.62	0.70	0.61	0.62	0.63	0.53	0.53	0.39
U^c	0.37	0.30	0.37	0.37	0.36	0.43	0.39	0.45
U	0.94	0.92	0.85	0.82	0.82	0.76	0.74	0.83
Univariate Model								
MSE	1,064	1,070	1,069	966	928	958	876	802
U^m	0.01	0.01	0.01	0.03	0.02	0.02	0.00	0.00
U^s	0.11	0.30	0.31	0.41	0.36	0.32	0.24	0.14
U^c	0.89	0.69	0.68	0.56	0.62	0.66	0.76	0.86
U	0.78	0.68	0.67	0.65	0.68	0.67	0.68	0.74
Sample size (n)	31	30	30	29	29	28	27	26

Note: Soybean—bushels, soybean oil—pounds, and soybean meal—tons.

Figure 2.1: Results of the StudyRausser and Carter

p. 495). Consecutive crop failures in Brazil in the 70's are responsible for the skewed forecasts of the model and the futures prices.

The inconsistency in the predictability of the model led the authors to perform further tests to test whether the model in the study performed better than the futures market. The MSE's of the futures market were compared to those of the ARIMA model in order to determine if they were statistically different. The test results concluded that the MSEs were in fact different for the two models. Therefore they were able to conclude that the futures market forecasts were inferior to those of the model, except for the soybean oil.

The study concluded that the multivariate and the univariate ARIMA models outperformed the futures markets for soybeans and soybean meal but not the soybean oil for both short run and long run. The study did not provide any empirical evidence for the cost and benefit condition. Though it mentions that they are running test with the ARIMA models by applying naïve buy and hold strategies to provide evidence that the sufficient condition also holds for the model. Thus the study provides support for the inefficiency of the futures markets for the soybean and soy meal markets.

The study performed by Rausser and Carter (1983) is important to this study because it provides methods to compare the results from the neural nets against those of the futures market. The MSE from the neural net forecasts can be compared with those of the futures market to find support for the inefficiency of the futures market. The Theil's inequality theorem will be an important one in this regard. Also, the exogenous and endogenous variables used in this study provide insight to the variables that can be used by the neural net to forecast the soybean prices.

2.2 Article 2: Forecasting Performance of Corn and Soybean Harvest Futures Contracts

The article by Kenyon et al. (1993), "Forecasting Performance of Corn and Soybean Harvest Futures Contracts" explores the price-forwarding performance of futures prior to planting season from a producer's perspective. The author claims that commodity futures prices prior to the season are not a reliable indicator of the harvest price. He argues that futures were a good indicator of the harvest price prior to 1973 and provides evidence for their progressive inability to do so afterwards. He relates the inefficiency of the futures markets post-1973 to improvement in storage facilities, government programs, yield calculation, and competition with international markets.

Producers make marketing plans prior to plantation of any crop by estimating the price they will receive at harvest time. In the case of corn and soybean, the planting season starts in April. As the author mentions, there is a lack of available information through which the producers can make well-grounded speculation for the harvesting price at that time. Any reliable data in relation to stock, yield, exchange rates, and such is available from the United States Department of Agriculture (USDA) in the form of a report entitled Agricultural Supply and Demand Estimates (ASDE). This report is published in the harvest expectation in either July or August. Consequently, the producers must rely on their own production and price estimates for the upcoming year. The failure of futures prices to predict harvest prices could, therefore, be expected to rise inefficiencies.

The author mentions that studies performed prior to 1973 by other econometricians suggest that the price at harvesting time should not exceed the price of the previous year's crop plus the storage costs. These studies also found that futures contracts were a reasonable forecast of the harvest price. Kenyon et al. (1993) found that the pre-1973 data was consistent with their assertions, although the post 1973 data was not. He attributes the structural break in the data to the change in the exchange rate market, as it moved from the Bretton Woods system to market determined post-1973. Under the Bretton Woods system, the government controlled the exchange rate and exports of commodities did not have to account for the exchange rate risk. Commodity markets were, for the most part, government controlled. Post-1973, the volatility of the exchange rate added to the risk in the commodities market. Government programs became more market oriented, and volatility in prices, yields, exports and stocks increased. The information available to the producers and the potential buyer didn't change, although the USDA made an effort to make information available in the form of ASDE.

Two models were used by the author to support his argument. The first model,

$$F_H = \alpha + \beta F_S \quad (2.11)$$

measures the ability of the spring price(F_S) of a harvest futures contract to forecast the harvest price(F_H) of the same contract. The second model was formulated to test the impact of stocks, yield forecasting ability, and government loan rates on price forecasting accuracy. The model is as follows,

$$FV = \beta_0 - \beta_1 SMPU - \beta_2 R2YLD + \beta_3 (PRICE - LOAN) \quad (2.12)$$

where,

FV - estimate of forecast variance

SMPU - April 1 stock minus the predicted use to harvest

R2YLD - R2 from yield forecasting equations

PRICE-LOAN - April 15 futures price minus loan rate

The author claims that independent variables are negatively related to the dependent variable, except β_3 . For SMPU, as the predicted use increases the demand for the harvest will increase. Hence, the store of stocks will have a lower impact on the prices at harvest time, and thus price from last years harvest plus storage is no longer a good indicator of the harvesting price. Therefore a decrease in stock will increase FV. Variability in yield estimates yield forecasting equations is a cause of uncertainty of the true yield at harvest time. Since R^2 is inversely related to yield forecasting accuracy, a low R^2 (and a high yield forecasting error) adds price forecast variance on the prices at harvest time. The government loan rate is effectively a floor on the prices at harvest time. Therefore, the closer the price is to the loan rate the less variance there can be, since the price will not generally go below the loan rate.

The data used in the study was taken from 1952 through 1991 of December corn and November soybean futures. The choice was based on the dates of maturity, as they are the first futures contracts to mature after harvesting.

The stacky first ran the F_H model on all the data 1952 - 1991. The OLS results for the period of the data gave high R^2 , yet the variance of the forecast error was

not constant therefore failing one of the OLS assumptions. Further, misspecification test results provided evidence of normality, heteroskedasticity, and mis-specified functional form. Thus theory suggests that there is a structural break in the data. Usually econometricians add a dummy variable to account for the difference in the data, but Kenyon decided to break the data into two periods based on the market changes. Two sub-periods, 1952-72 and 1974 - 91 were chosen. The year 1973 was left out because this year was considered a transitional year of the changes in the market. The observation from the first period where further reduced by four years recursive coefficient plots which suggested that β and the standard deviation started to change after 1968. So the first period was reduced to 1952-68. The author ran regressions on these two periods even though there were less than 32 observations, which is generally accepted as the minimum number of observations where normal distribution can be assumed. He performed influence diagnostics to test the sensitivity of the results to outliers, to make up for the loss in observations. The results were supportive of the author's argument. In the first period the spring futures prices of the harvest contract explained at least 70% of the price variation at harvest for both corn and soybean. In the second period the spring futures prices of the harvest contract explained only 4%, and 15% of soybean and corn respectively, and the β coefficients were all insignificant. The influence diagnostic test results for the first period supported the conclusion of the OLS results. After removal of the outliers in the diagnostic tests, the β and the intercept changed, but the equation still suggested that the spring futures prices of the harvest contract explained much of the variance in harvest prices. The diagnostic tests for post-1973 were different for both soybeans

and corn. The soybean diagnostic test results did not affect the results as the new β were still statistically insignificant. The corn diagnostic tests were opposite to that of soybean. The removal of outliers improved the overall significance of the model, and increased the R^2 . The highest R^2 was achieved by removing 1974, 1980, and 1981 giving an R^2 of 0.43 as opposed to 0.15 with the outliers, as pointed out by Kenyon et al. (1993) . In conclusion the results for post-1973 were not nearly as significant as pre-1973.

Testing of the second model used the logarithm of recursive and window estimated of ten-observation window size of FV. It also added the squares of SMPU and PRICE-LOAN to the model because the relationship between these two independent variables and the dependent variable was not considered to be linear. The results of the second were consistent with the goals of the study. The signs of the coefficient were as expected. Both β_2 and β_3 were statistically significant, but β_1 unfortunately was not. The R^2 also for the corn and soybean were 0.74 and 0.86 respectively. Kenyon et al. (1993) found autocorrelation in this model, but did not correct for it.

The study performed by Kenyon et al. (1993) does an adequate task of pointing out the inefficiency problems related to futures as forecasters for the harvesting price. His theoretical approach is sound, though his empirical testing could be improved. The first model only contains one independent variable, April future contract prices. He, himself, points out that more factors affect the harvest price post-1973, yet he doesn't account for them. For the second model, results from the regression were presented without correcting for autocorrelation. Thus the R^2 and t-stat could be artificially inflated, and thus there could be fear for type II error.

From the study it can be concluded that spring futures contracts are not a good indicator of the harvest price of both corn and soybean. Thus using soybean spot price indicators it could be possible to out perform the futures market in predicting the futures spot price using artificial neural networks. If the futures market can be outperformed then it is possible to make profit in the futures market by evaluations of the predictions of the future spot price from the neural network.

2.3 conclusion

The literature reviewed suggests that the futures market is inefficient in correctly pricing the futures contracts. This would suggest that if an appropriate model and a suitable neural network can predict the future spot prices more accurately than the futures then the relative accuracy condition, one part of inefficiency, can be shown. This relative accuracy condition suggests that a model exist whose forecasts are more accurate than the futures contracts. The other condition of sufficiency, discussed by Rausser and Carter (1983) can be achieved through the analysis of the cost/benefits of using neural networks in prediction. The sufficiency condition for neural network will be discussed in the methodology section.

CHAPTER 3

Methodology

Introduction

As Hill et al. (1996) mentions that in the last few decades, there has been a lot of research that has been directed at predicting the market. There have been studies performed on the efficiency of commodity futures markets using traditional statistical techniques for time series forecasting such as the one performed by Kenyon et al. (1993). Having exhausted the traditional statistical techniques researchers have looked else where especially in the realm of artificial intelligence, a field of computer science. Research in artificial intelligence has been aimed at creating software and hardware to mimic the human behavior. This research has given rise to artificial neural networks which are modelled on the human brain. This section will present the basics of neural net architecture, and the types of nets this study will use to predict the future spot prices of soybeans.

3.1 Development of Artificial Neural Networks

Artificial neural networks (ANN) have given a new meaning to artificial intelligence (AI). Previously AI was mostly limited to sequential processing and only certain representations of knowledge and logic. The new technology, artificial neural networks, have allowed computers to mimic the processing characteristics of the human brain thereby giving computers the capability of recognizing patterns based on experience. The development of ANN was inspired by research performed on the processing of information in the human brain. Major development and research on ANN was performed during the 1990s (Trippi and Turban, 3).

The ANNs were inspired by research performed on the human brain. The human brain comprises of approximately 100 billion neurons. These neurons are unique from other cells in the body because they don't die. Other cells in the body reproduce and thus replace themselves and die off. The longevity of the neurons in the brain might help explain why humans retain information. Neurons in the brain are grouped together in networks, and therefore the brain can be looked as a group of neural networks (Trippi and Turban, p. 4).

All the activities in humans related to information processing resulting to intelligent behavior are carried out by the brain. The brain and other neural networks in the human body make up the central nervous system. This central nervous system allows humans to process stimuli in the environment around them. This recognition and processing of stimuli by the central nervous system gives humans the ability to learn and react to changes in their environment (Trippi and Turban, p. 4).

The human nervous system is staggeringly complex. “Knowledge about the central nervous system’s overall operation is so limited that there is little to guide those you would emulate it”(Wasserman, p. 11). Artificial neural networks try to mimic just a few basic functions of the neurons and the neural networks.

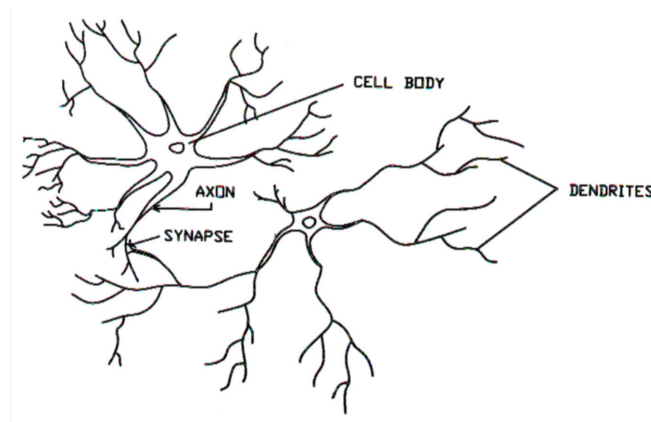


Figure 3.1: A biological neural network of two neurons (Wasserman, p. 13)

Figure 3.1 shows a network of two neurons. The neuron is made up of the cell body which evaluates and processes signals received from the dendrites. As figure 3.1 illustrates, the dendrites are the connections between two neurons. Dendrites extend from the cell body to other neurons where they receive signals at a connection point called a synapse. These signals are transmitted from the dendrites to the cell body where they are processed. These signals or inputs “are summed, some inputs tending

to excite the cell, others tending to inhibit its firing”(Wasserman, p. 12). If the inputs excite the cell body then the output of the neuron is carried through the axon and delivered to the synapse where it is again processed and then transmitted as input for the second neuron. Information in the form of signals produced by the first neuron can be transmitted unchanged, or they can be manipulated before they are transmitted by the synapse. “The synapse is able to increase or decrease its strength of connection and causes excitation or inhibition of a [signal from the] subsequent neuron”(Trippi and Turban, p. 5).

In the manner described above neurons in a biological network communicated and process signals with each other. These networks take inputs from the environment or other networks and create an output. The information about different signals stored in the nucleus of the neuron help the network to learn and react to different signals. Research in AI aimed at mimicking this behavior led to the development of artificial neural networks. These networks similar to neural networks in the brain are designed to learn and adapt to signals through a process of error minimization. Analogous to learning performed by a human child.

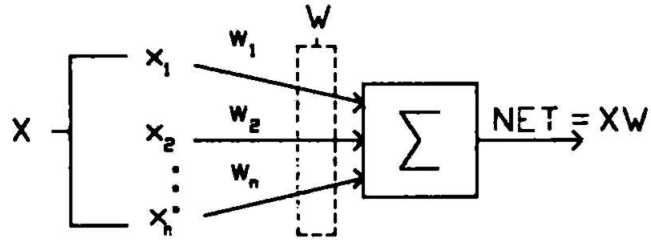
3.2 Components and Structure of Artificial Neural Networks

3.2.1 A Simple Neuron

The ANN consists of artificial neurons (*also known as neurodes*) which are the basic processing elements of the network. The artificial neurons process signals received

and transmits them to the other neurons much the same way as their biological counterparts. “The artificial neuron was designed to mimic the first-order characteristics of the biological neuron”(Wasserman, p. 13). Each neuron receives and input from another neuron from a previous layer or from outside the network. Every input is multiplied by a corresponding weight much like the synaptic strength of a biological neuron. After multiplication all the values are added up to determine the activation level of the neuron. A simple artificial neuron without an activation function can be seen in the figure 3.2. In figure 3.2, the variable X corresponds to a vector of inputs $x_1, x_2, \dots x_n$. Each of these inputs are multiplied by an associated weight with the weights labeled $w_1, w_2, \dots w_n$. Each of the “weights correspond to the ‘strength’ of a single biological synaptic connection”(Wasserman, p. 13). Similar to the input vector X , the weights are collectively labeled as vector W . The result from the multiplication of the input vector with the weight vector is given to the summation block analogous to the cell body. In the summation block all the results are added up to give the output labeled ‘NET’ in figure 3.2. Sometimes a constant can be added to the output of the summation block. This constant is known as the bias or the ‘firing threshold’. Later in the section the significance of the threshold will be introduced.

The result from the summation block, NET is yet processed further. Figure 3.3 shows a neuron with an activation function. As figure 3.3 illustrates function F is applied to the NET to give the output of the neuron OUT. “Based on this level, the neuron may or may not produce an output”(Trippi and Turban, p. 9). The relation between the NET and the OUT may be linear or non-linear depending on the type of activation function used.



$$\text{NET} = x_1 w_1 + x_2 w_2 + \dots + x_n w_n$$

Figure 3.2: An Artificial Neuron (Wasserman, p. 14)

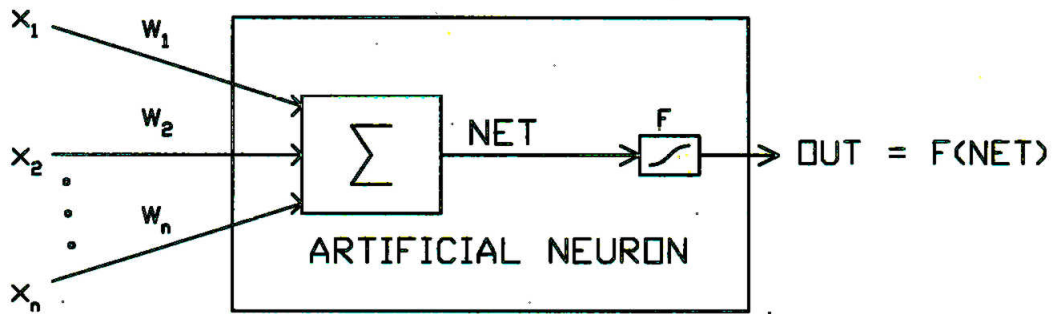


Figure 3.3: An Artificial Neuron with activation function (Wasserman, p. 15)

The mathematical model of a neuron can be represented as follows:

$$y = f\left(\sum_{i=1}^n w_i x_i - \tau\right) = f(\underline{w} \cdot \underline{x} - \tau) \quad (3.1)$$

where y is the output OUT, the summation in the braces is equal to NET, and τ is the firing threshold. The threshold can be either positive or negative depending on what kind of application the neuron is emulating. f represents the activation function and the $\underline{w}$ and $\underline{x}$ are the vectors of weights and inputs respectively.

3.2.2 Activation Functions

There are several different kinds of activation functions that are used in neurons. The purpose of the activation function is to modify the NET levels to a reasonable value. Without an activation function the output from the summation block, NET can be very large. Thus the activation function is used to normalize the output NET. Figure 3.4 shows four kinds of activation functions. These four activation functions can be used to create many different kinds of networks. The selection of the activation function depends on the network's operation.

The activation functions shown in figure 3.4 are linear, hard-limit, log-sigmoid, and tan-sigmoid. The linear activation function calculates the neuron's output by simply returning the value passed to it from the summation block. By using this type of function a neuron can be trained to find linear approximations from its inputs to its outputs.

The hard-limit activation function shown in figure 3.4 limits the output of the summation block to either 0 or 1. This kind of function is called a *threshold detector*. For example, any input value from the summation block, NET above a constant number is changed to 1 and the values below the constant to 0. This enables the neuron

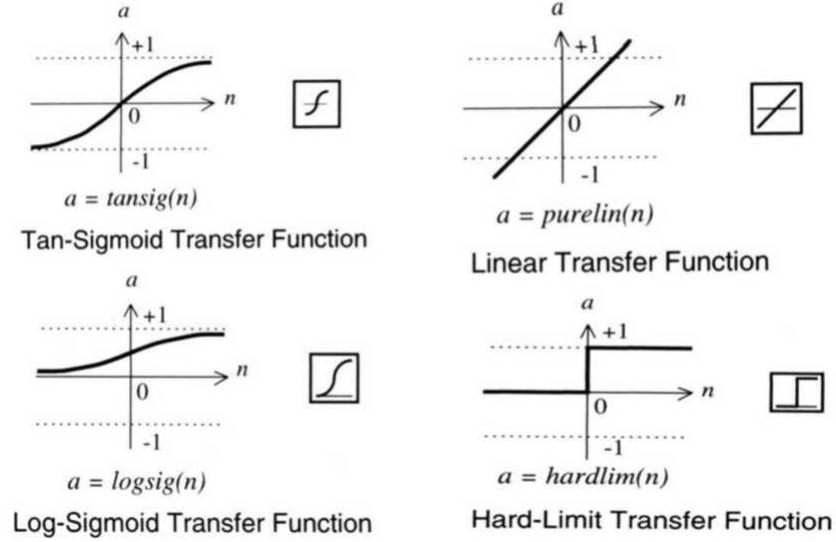


Figure 3.4: Activation Functions (Demuth and Beal)

to differentiate the input vector into two distinct regions. This kind of activation function is used in preceptrons.

The two activation functions important to this study are the log-sigmoid and the tan-sigmoid functions. The log-sigmoid function generates outputs in the range and including 0 and 1 as the neuron's inputs range from negative to positive infinity. The tan-sigmoid function on the other hand generates outputs in the range and including -1 and 1 for inputs ranging from negative to positive infinity. The mathematical representation of the log-sigmoid function is as follows:

$$\gamma_T = \frac{1}{1 + e^{-y}} \quad (3.2)$$

The tan-sigmoid function is as follows:

$$\gamma_T = \tanh(y) \tag{3.3}$$

As mentioned earlier the activation function compresses the range of the output from the summation block such that it never exceeds some low limit. Therefore the activation functions especially the sigmoid functions are known as ‘squashing functions’. “Multilayered networks have greater representational power than single layered networks only if non-linearity is introduced, and the squashing function produces the nonlinearity” (Wasserman, p. 45). The layers and the structure of the neural net will be introduced in the next section.

3.3 Feedforward Networks

There are many different kinds of neural nets some of which are called perceptrons, feedforward, and hopfield networks. Analyzing all these networks and their applications is beyond the scope of this study. Only the feedforward network will be discussed.

A simple two layered feed forward network can be seen in figure 3.5. The inputs to the network are received through the left by the input layer and the output of the network are generated on the right by the output layer. The feed forward networks is built by three types of processing units: input, output, and hidden. The input units make up the input layer as the output and the hidden units make the output and hidden layers respectively. The input layer accepts signals from the environment and thus resides in the lowest layer of the network. The input layer has no function other

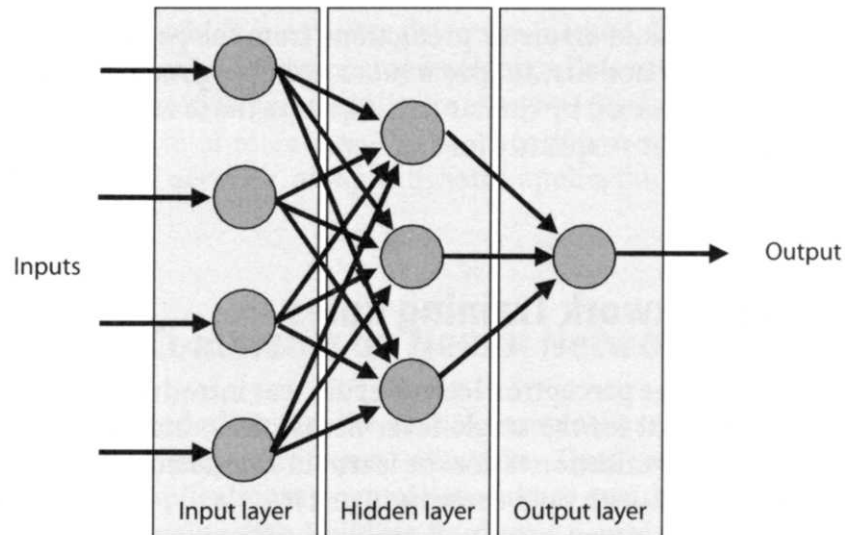


Figure 3.5: Feed Forward Neural Network (Shadbolt and Taylor, p. 91)

than transmitting the signals environment to the next layer of the network, which in the case of figure 3.5 is the hidden layer. The hidden and the output layer are made of neurons such as the one described earlier in figure 3.3.

The nodes or neurons in the hidden layer of the network are first processing nodes of the network. The nodes in the hidden layer do not interact with the environment and thus referred to as ‘hidden nodes’. Output from the hidden layer are transmitted to the next layer which in the case of figure 3.5 is the output layer. The output layer generates signals out to the environment. This layer performs calculations just as the hidden layer. The network in figure 3.5 is called a two layered network since there are two processing layers in the network because the input layer performs no calculations.

If a network only has an input and an output layer then it is called a single layered network.

The feed forward networks learn through adjusting their connection weights. It is not easy to define weights for the connections to the inputs from the environment to the expected outputs. There are many algorithms that are used to adjust the weights of the neural networks in order to learn the relationship between the inputs and the outputs. One of the widely used algorithm for the training of feed forward neural networks is the backpropagation learning algorithm.

3.3.1 Back-Propagation Learning Algorithm

A back-propagation algorithm is one of type of training algorithms used to train feed forward networks. “The learning rule now known as the ‘back-propagation’ is a generalization of the Widrow-Hoff rule for multi-layer networks” (Davalo and Naïm, p. 47). The rule propagates the error, calculated from the output of the network and the actual output, backward through the intermediate layers towards the lowest layer of the network, the input layer. This propagation is analogous to the propagation of the input from the input layer to the output layer through the intermediate layers of the network.

The training of the feed forward neural network using the back-propagation algorithm is performed by providing the network with a set of the inputs and a corresponding set of outputs. For example, this set of inputs and outputs can be related to predictors of prices of a commodity as inputs and the price of the commodity as outputs. Thus the network takes the inputs and using the current weights in the

network produces a corresponding output. This phase is called forward propagation. In the next step total mean squared error(MSE) is calculated between the outputs from the net and the actual output of the training set. This MSE is then propagated backwards from the output layer towards the input layer each time used to adjust the weights between each of the subsequent layers of the network.

3.4 Training Neural Networks

There are two ways to calculate train a network: incremental and batch training. In incremental training the error is calculated after each input is presented to the network and an output is generated. In the batch training mode a full data set is applied to the network and then the error is calculated for the full data set and propagated backwards through the layers of the network. Each pass over the full data set is known as one *epoch* Demuth and Beal, p 5-9.

In the fashion indicated above the network is trained for certain number of epochs until the mean squared error is below a particular threshold. In practice, Shadbolt and Taylor (2002) states that there are three kinds of data sets: training, validation, and the test set. The test set and the validation set are stochastically different but share the same distribution. The validation set is usually randomly selected from the training set and the observations are removed from the training set. The particular halting threshold mentioned earlier is obtained by running the neural network on the validation set and then comparing the movement of MSE of the validation and training set over time. Figure 3.6 illustrates the MSE of the training set in the graph

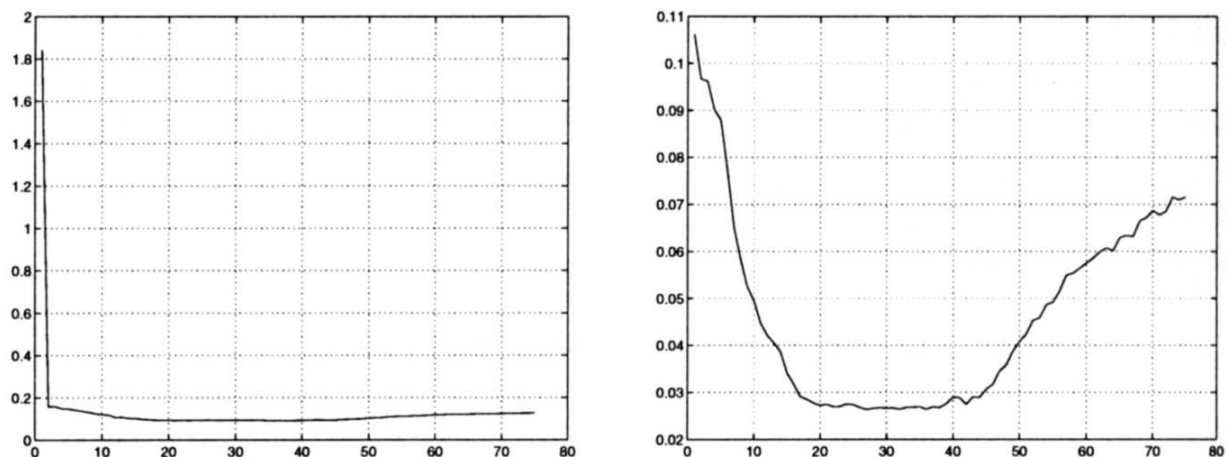


Figure 3.6: Training and Generalization Errors Showing Over Fitting (Fine, p. 244)

on the left and the MSE of the validation set in the graph on the right. The x-axis of the graphs represent number of epochs and the y-axis represents the MSE. As the number of epochs increases, “the validation error generally decreases, but there comes a time when this error starts to generally increases even though the training set error is still decreasing” (Shadbolt and Taylor, p. 244). Thus from the increase in the MSE of the validation set it is judged that over-fitting of the data set is taking place by the neural network. By over fitting it means that the neural net is memorizing the training data set therefore loosing its power to generalize the relationship between the inputs and the outputs of the validation set. Over fitting of the data by the network would signify that the net will predict poorly when it is presented with the test set.

3.5 Functional Approximation Power of ANN

In general neural nets are considered to be able to approximate any function given that a certain number of nodes are present in the network. “The first clear insight into the versatility of neural networks for use in functional approximations came with the discovery of Kolmogorov’s Mapping Neural Network Existence Theorem”(Hecht-Nielsen, p. 131). Hecht-Nielsen (1990) states that, “At the time(1986) it was interesting, because no other similar neural network performance guarantee had been found”(p. 124). Later other theorems where developed which showed that neural networks could approximate any function of interest. One such theorem, back-propagation approximation theorem, shows that the back-propagation network[†] is able to implement any function to any desired degree.

Back-Propagation Theorem states that, “*Given any $\epsilon > 0$ and any L_2 function $f : [0, 1]^n \rightarrow \mathbf{R}^m$, there exists a three-layer back-propagation neural network that can approximate f to within ϵ mean squared error accuracy*”(Hecht-Nielsen, p. 132).

To understand the theorem above some background information is provided by Hecht-Nielsen (1990). He states that, for any given square integrable function $g : [0, 1]^n \rightarrow \mathbf{R}$ (i.e., any function for which $\int_{[0,1]^n} |g(\mathbf{x})|^2 d\mathbf{x}$) it can be shown by Fourier series that the series

$$\hat{g}(\mathbf{x}, N) = \sum_k c_k e^{(2\pi \mathbf{k} \cdot \mathbf{x})} \equiv \sum_{k_1=-N}^N \sum_{k_2=-N}^N \dots \sum_{k_n=-N}^N c_{k_1 k_2 \dots k_n} e^{(2\pi i \sum_{q=1}^n k_q x_q)} \quad (3.4)$$

[†]This is another name for a feed forward network using back-propagation learning algorithm

where

$$c_k \equiv c_{k_1 k_2 \dots k_n} = \int_{[0,1]^n} g(\mathbf{x}) e^{(-2\pi i \mathbf{k} \cdot \mathbf{x})} d\mathbf{x} \quad (3.5)$$

converges to g in the sense that

$$\lim_{N \rightarrow \infty} \int_{[0,1]^n} |g(\mathbf{x}) - \hat{g}(\mathbf{x}, N)|^2 d\mathbf{x} = 0 \quad (3.6)$$

Through the example above Hecht-Nielsen (1990) shows that the integral of the square of the error of approximation of one function by another go to zero. “This means that the approximation of any function can be achieved to any desired accuracy in the MSE sense. For a given function $f : [0, 1]^n \rightarrow \mathbf{R}^m$, f is L_2 if each of f ’s coordinate functions is square-integrable on the unit cube. Considering functions like f and assuming that the $\mathbf{x}$ vectors are chosen uniformly, then if a vector function of a vector variable is in L_2 then each of its components can be approximated by its Fourier series to any desired degree of accuracy in the MSE sense” (Hecht-Nielsen, p. 132).

Through the explanation above Hecht-Nielsen (1990) shows that any real world problem can be approximated by a three layer back-propagation network. Hecht-Nielsen (1990), states that even though the theorem proves that a three layered network is always enough to model any real world function, it is impractical to try and model it with a three layered network due to the impractically large number of neurons required in the hidden layer to do so. These problems can be solved through simpler nets using more than three layers in their networks.

One draw back that Hecht-Nielsen (1990) points out is that the theorem above proves that any real world problem can be approximated through a three layered

back-propagating network, however it does not prove that the correct weights needed to solve that problem can be learned through any existing learning law. This suggest that there exist a three layered network to solve the real world problem, but finding and training it is the challenge.

3.6 Using Neural Networks

Today, ANNs are used for classification, clustering, and pattern recognition of data. Application have been developed using ANNs that forecast stock and commodity prices, bond ratings, foreign exchange rates, T-bills, bonds, and inflation. ANNs have also been found helpful for security trading systems, pricing and hedging of derivatives, forecasting the financial success and demise of firms and predicting merger targets. Banking sectors has used neural networks to predicting credit risk, evaluating signatures, voice recognition systems, and finger print recognition(Krishnaswamy et al., p. 75).

3.6.1 Strengths and Weaknesses of ANN

As shown in the previous section artificial neural networks have the capability of approximating any real world problem. Davalo and Naïm (1991) states that neural network do not require knowledge to be formalized, they are appropriate to domains in which knowledge is scanty. This means that a formal model between a input and output variables is not necessary to predict the relationship between them. The neural network has the ability to adjust so that the impact of the variable that do

not have any relationship with the dependent variable are not weighted as heavily. Secondly ANNs can generalize a relationship between input and outputs if the data presented is noisy, incomplete, or previously unseen. Also an ANN can be retrained to learn new relationships thus showing a high degree of flexibility as well.

Neural Networks also have their weakness. For one ANNs do not have any justification for their results as the output is generated through the intricate valuation of the weights in the network. With current technologies the training to train a ANN can long and tedious. To get a good prediction out of the ANN is difficult since it depends on many factors such as the “The best way to represent input data and the choice of architecture is still mostly subject to trial and error”(Trippi and Turban, p. 13). Also neural networks require lengthy computing and training times which may not be suitable for some problems.

With neural network there is also no guarantee of an optimal solution to a problem, or a completely certain solution. They also don't have repeatability of outputs given the same data. None-the-less, “trained neural networks can be used to make good classifications, generalization, or decision.

3.6.2 Feasibility of Using ANNs

Today software to build and train artificial neural nets is easily available. In the past specialized hardware that performed neurocomputing were extremely expensive and hard to maintain. One such computer was the Electrically Trainable Artificial Neural Network (ETANN) was developed by Intel. Today software for personal computers is available to build and run neural networks.

NeuralWare is one of the companies that produces relatively extensive software for neural computing. Their prices range from \$100 to \$15,000 dollars with maintenance due annually of about \$300 to \$3,000. More common software can be found that can be added on to existing statistical software like Excel and Lotus. The Palisade Corporation produces a software called Braincel that can be added on to Excel. The software costs about \$250 and uses a version of back-propagation called backpropagation, developed by Palisade. Other software can be downloaded from the internet some of them are shareware and some free (Krishnaswamy et al., p. 78).

3.6.3 ANNs in Financial Forecasting

A key element in improving forecasting ability in financial markets is the acquisition of superior data that the average investor doesn't have. Acquisition of such data is hard. Even if superior data is available still "as soon as transaction yielding excess returns are exercised, the market renders the information public, thereby eroding its value" (El Shazly and El Shazly, p. 346). Thus the choice then remains in processing of current data more efficiently in order to get an edge on the average investor.

The success of ANNs in the financial world are evident, but still there is relative debated on whether they can be classified as a non-parametric statistical approach or just another computer technique. Since the basis of neural nets reside in neurology and computer science a judgement on either classifications can be hard to make. Krishnaswamy et al. (2000) suggests that there is recent literature that supports that ANNs can be classified as a statistical technique because they can be classified as a learning technique and can be demonstrated to be technically equivalent to a series

of recursive nonlinear regressions. For the scope of this study ANNs will be used as a statistical technique to predict the future spot prices of soybeans(Krishnaswamy et al., p. 76).

CHAPTER 4

Empirical Results

Introduction

This chapter focuses on achieving the goal of this study, mentioned in the introduction and its background presented in the theory and methodology chapters, through empirical analysis. The goal restated, is to check the efficiency of the soybean futures commodity market. The market will be tested for the weak form of efficiency outlined by Fama (1970) in the efficient market hypothesis (EMH). According to the weak form of the EMH, past prices, volume, and other market statistics provide no information that can be used to predict future prices.

Superior information about the price changes in the future are hard to get. Even if superior information is used, as soon as market transactions are made on that superior information, the information is regarded as public and profits are diluted. Therefore, another approach to attain an edge on the market is considered which is efficient processing of publicly available information.

Artificial neural networks are used to efficiently process the publicly available

information, and hence predict the future price of the soybean commodity. Statistical analysis of market trends are relatively old, and other measures such as Expert Systems have limited capability in predicting market behavior on information that the system has not seen before. As shown in previous chapters artificial neural networks can prove to be an important tools in forecasting markets do to their power of generality.

In this chapter results of empirical testing performed using neural networks to predict future spot prices of soybean commodity will be presented. Three sets of data were used to predict the future spot prices of soybeans. The forecasts of the neural network model were compared with those of the futures. A trading rule was also developed to show if profits can be made using the information provided by the neural network.

4.1 Model Specification

To check for the efficiency of the futures commodity market neural networks where developed to predict the future spot price using three different sets of input variables. Most of these variables were discussed in the theory section. The hypothesis being tested by this study is as follows:

The futures commodity market is not considered to be weak form efficient as per the EMH hypothesis presented by Fama (1970) because technical analysis through neural networks of past information such as prices of futures contracts, prices of soybean

commodity and its compliments, and other factors affecting future spot price can be used to predict future spot prices.

To empirically test this hypothesis a trading rule has to be made using the forecasts made by the neural networks. Market efficiency as states by Keane (1983) is totally an empirical issue. Thus, a market can be considered inefficient for the data and the commodity used if money can be made consistently through the analysis of the past information with neural networks.

4.1.1 Measurement of Operationalized Variables and Models Tested

While working with neural networks operationalized models are not required. Due to the black box nature of neural networks the effect of any independent variable on the dependent variable/s cannot be determined. Therefore, three conceptual models were developed. The relationship of these variables was defined in the theory chapter. In this section the variables and their measurements will be discussed. The three models used in this study are as follows:

Model I:

$$\Delta SP_{1m}, \Delta SP_{2w}, \Delta SP_{1w} = f(T, SM_N, SM_{T_N}, SM_{N+1}, SM_{T_{N+1}}, SM_{N+2}, SM_{T_{N+2}}, SO_N, SO_{T_N}, SO_{N+1}, SO_{T_{N+1}}, SO_{N+2}, SO_{T_{N+2}}, SB_N, SB_{T_N}, SB_{N+1}, SB_{T_{N+1}}, SB_{N+2}, SB_{T_{N+2}}, CF_N, CF_{T_N}, CF_N, CF_{T_N}, HS, CS, SOI, SMOI, SMV, SBOS, SOOI, BOV, IR, SBSIL, SBSOH)$$

Model II:

$$SP_{1w}, BA = f(T, SM_N, SM_{T_N}, SM_{N+1}, SM_{T_{N+1}}, SM_{N+2}, SM_{T_{N+2}}, SO_N, SO_{T_N}, SO_{N+1}, SO_{T_{N+1}}, SO_{N+2}, SO_{T_{N+2}}, SB_N, SB_{T_N}, SB_{N+1}, SB_{T_{N+1}}, SB_{N+2}, SB_{T_{N+2}}, CF_{N+1}, CF_{T_{N+1}}, CF_{N+2}, CF_{T_{N+2}}, HS, CS, SOI, SMOI, SMV, SBOS, SOOI, BOV, IR, SBSIL, SBSOH)$$

Model III:

$$SP_{1m} = f(T, SB_N, SB_{T_N}, SP)$$

In the first model the variables being predicted by the neural network were the change in spot price of soybeans one month(ΔSP_{1m}), two weeks(ΔSP_{2w}), and one week into the future(ΔSP_{1w}). These variables are also referred as the dependent variables because they are being determined endogenously by the model. These variables were generated by lagging the spot price of soybeans in the Ohio market($SBSOH$).

The first independent variable is time of year, T . Since all the observations in the data are recorded on daily bases therefore this variable keeps count of the day of the year. Soybean is a seasonal commodity as described in the theory section. Therefore the time variable allows the neural net to capture seasonal patterns between between the independent variables and the dependent variable.

The next 22 independent variables are the futures contract prices and the days left till their expiration of the important commodities that are used to predict spot prices of soybeans in the future. The commodity contract chosen are soybeans, soybean meal, soybean oil, and corn. Only a few contracts were chosen for each of the commodity. Soybean commodity has 7 contracts expiring in a given year in the months of January, March, May, July, August, September, and November. To have consistency in the data only January, May, and November contracts were chosen. Therefore the current contract to expire SB_N would start with approximately 120 days till expiration until the next but one contract to expire SB_{N+1} became the current contract and the SB_{N+2} became the next but one contract to expire. In this way the neural net had information about the next three contracts to expire predicting

the spot prices at exactly three points in a given year. The soybean meal and oil have more contracts expiring in a given year than the soybeans. The months of expiration for soybean meal and oil are January, March, May, July, August, September, October, and December. Since the soybean meal and oil spot prices both define the soybean spot price as shown in the theory chapter therefore the exact same months used for soybeans were used for soybean meal and oil. Corn is a substitute for soybeans in consumption as stated in the theory chapter thus it was an important variable to be added to the model. Since there is no January contract that expires for corn thus only May and November contracts were used. The list of the variables relating to futures contracts listed in the model and what they measure is as follows:

- SM_N - is the price of the next expiring contract for soybean meal. The price for soybean meal is quoted in US dollar and cents per ton. One contract of soybean meal trades 100 tons (2,000 pounds/ton) of soybean meal. The last trade on these contracts can be made at the seventh business day preceding the last business day of the delivery month. The strike price increment of the soybean meal contract are \$5/ton for futures priced below \$200/ton and \$10/ton for futures priced above \$200/ton. Strike increment means the amount by which the price can increase above the strike price[†]. So if the contract is priced below \$200/ton then with the next trade the price of the soybean contract will only increase by \$5/ton. As mentioned in the theory chapter, trading will stop

[†]The strike price is the price at which the holder of a futures contract can buy or sell the underlying commodity on or before the expiration date

if the price of the contract reaches the price ceiling for that contract by the Commodity Exchange.

- SO_N - is the price of the soybean oil contract for the next expiring contract. The price of soybean oil contract is calculated in US dollars and cents per pound. Soybean oil trades for 60,000 pounds of crude oil per contract. The last trading day is the seventh business day preceding the last business day of the delivery month. The strike price increment for soybean oil is $\frac{1}{2}$ cent/pound for futures priced below 30 cents per pound and 1 cent/pound for futures priced above 30 cents/pound.
- SB_N - is the price of the next contract to expire for soybeans. The soybean is quoted in US cents/bushel. The contract trades 5,000 bushels of soybeans. The last trading day for the soybean contract is the same as the soybean meal and oil contracts, which is the seventh business day preceding the last business day of the contract month. The strike price increments are 25 cents/bushel.
- $SM_{T_N}, SO_{T_N}, SB_{T_N}$ - measures the days left until the expirations for the SM_N, SO_N, SB_N contracts respectively. These values as explained before range from zero to about 120 counting the days till expiration of the contracts.
- $SM_{N+1}, SO_{N+1}, SB_{N+1}$ - are the prices for the next but one contracts to expire for soybean meal, soybean oil, soybeans. These contracts will be expiring May if the next contract to expire for soybean meal, soybean oil, and soybean is January.

- $SM_{T_{N+1}}, SO_{T_{N+1}}, SB_{T_{N+1}}$ - measures the days left until the next but one contract expires for soybean meal, oil, soybeans. Usually this value ranges from about 120 to 240 for the soybeans and soybean meal and oil.
- $SM_{N+2}, SO_{N+2}, SB_{N+2}$ - This is the price of the next but two contracts to expire for soybean meal. If the next contract to expire for soybean meal is January, then the next but one contract will be May, and this contract, next but two, will be the contract that expires in November.
- $SM_{T_{N+2}}, SO_{T_{N+2}}, SB_{T_{N+2}}$ - measure the days left until the next but two contract expires for soybean meal, soybean oil, and soybeans respectively. The value of these variables ranges from about 240 to 360.
- CF_N, CF_{N+1} - is the price of the corn futures contract for the next expiring contract and the next but one expiring contract. Due to data restrictions only two contracts were chosen for corn to match two months of the soybeans May and November. Thus the labels for corn are a little misleading. The next but one contract(CF_N) for corn will be May if the next contract to expire for soybeans(SB_N) is January, and the next but one contract to expire for corn, CF_N , will be November if the next contract to expire for soybean is May. So both the variables for the corn futures contract switch between price of the May and November contracts. The price of the corn contract is quoted in US cents/bushel. The contract trades 5,000 bushels of corn. Similar to the soybeans contract, the last trading day for the corn contract is also the seventh business

day preceding the last business day of the contract month. The strike price increments for corn are 10 cents/bushel.

- $CF_{T_N}, CF_{T_{N+1}}$ - is the days to expiration for the next and next but one contract of corn futures. The days till expiration for the next contract range from 0 to 240 days, and the days till expiration for the next but one contract for corn range from 120 to 360 days.

The other commodities that were important in predicting future price of soybeans which are in the model are the spot prices of Corn, Hogs, Cattle, Soybean Meal, Soybean Oil, and prices of soybeans from two major markets in the US, Chicago and Toledo. The Corn spot price, CS , is the central Illinois bid price which is quoted in US cents/bushel. The hog spot price, HS , is the Omaha average hog spot price per day calculated as US dollars per 100 pounds. The cattle price is the price of Omaha slaughtered steer quoted in US dollars/100 pounds. The soybean oil price $SBOS$ is quoted in US cents/pound in the Decatur bid market. Due to data availability the soybean meal spot could not be included in the model. The spot of soybean included in the model are $SBSIL$ and $SBSOH$ for the Chicago and Toledo markets respectively. The price for soybean spot in both these markets is quoted in US cents/bushel.

Five other independent variables in the first model are related to the volume and open interest of the soybeans, soybean meal, and soybean oil futures. Due to data restriction the volume of futures traded of soybean oil could not be attained. The variables SOI , $SMOI$, and $SOOI$ measure the open interest of soybean, soybean meal,

and soybean oil respectively. The variables BOV and SMV measure the soybean futures trading volume and the soybean meal trading volume respectively.

The last variable of interest in the model is IR which is the 30 day interest rate on US treasury bills. This variable is measures the interest rate paid on borrow money. As mentioned in the theory section arbitrage opportunities depend on the cost of borrowing the cash to carry out the arbitrage. The cost of borrowing which is the interest rate paid on the amount that is borrowed is measured through this variable.

The same independent variables were used to predict the dependent variables of Model II. The dependent variables in model II were the future price one week from now, and the change in basis one week from now. The lagged Toledo price was used for these dependent variables as well. The futures contract used to calculate the basis was the next expiring soybean futures contract, SB_N , lagged one week same as the Toledo spot price for soybeans.

The last model, model III, is simpler model derived from the larger models. This model was used to predict Toledo market soybean spot price one month in the future. The independent variables used for this model are time(T), next expiring futures contract price SB_N , days left to the expiration of this futures contract (SB_{TN}), and the current Toledo spot price for soybeans(SP).

4.2 Data

The data for the study was collected from the DRI database published by Standard & Poor's. The data is of daily values for each of the variables of interest. The days

where the markets was closed were omitted from the data set because no values for the variables where available. As mentioned before only three contracts from soybeans, soybean meal, and soybean oil were chosen such that they would expire in the same month and seasonality of the price movements between these three variables can be ascertained by the neural network.

In total 4,631 days of data where collected spanning from 1980s to 2004. Observations from the early years of the 80s was eliminated because the data was absent for some of the variables till 1985. Similarly data was omitted for the year 2004 because data was not available for some of the recent days and therefore the data of the next but one, and next but two variables of commodity futures could not be created. With 35 variables over 4,631 days totaled about 162,085 observations on which to train and test the predictability power of the neural networks.

For the first two models sample data from the 80s, 90s, and 00s was used to train and forecast the commodities market. There was a limitation of 10,000 elements for the training and forecasting set in the software used to build these neural networks. Therefore only 250 trading days where used to train the network and it was tested on 80 trading days worth of data. After removing all the holidays from the data set, trading days in one year total approximately 249 days. Thus the neural net was trained on one year worth of data and tested on 4 months worth of data. Another fourth set was created from the data starting from 2000. The training data consisting of 270 trading days was randomly selected from 751 trading days between 2000 and 2002. Testing of the data was performed on 92 trading days in 2003. From the training data sets observations were randomly selected and removed to create validation data

sets which consisted of 30 trading days. The summary of the training and testing data sets for model one and two are given in table 4.1.

Training and Forecasting Data Set for Model I & II								
	1980s		1990s		2000		3yr Random	
	Training Set	Forecasting Set	Training Set	Forecasting Set	Training Set	Forecasting Set	Training Set	Forecasting Set
From	Day 2, 1980	Day 363, 1986	Day 4, 1993	Day 363, 1993	Day 2, 2001	Day3, 2002	Day 3, 2000	Day 2, 2003
To	Day 360, 1980	Day 112, 1987	Day 362, 1993	Day 112, 1994	Day 2, 2002	Day 119, 2002	Day 365, 2000	Day 134, 2003
Total # of Days	250 Days	80 Days	250 Days	80 Days	250 Days	80 Days	270 Days	92 Days

Table 4.1: Sample Data Sets for Training and Testing of Models I & II

Almost all the data was used in different sets to train and test the third model. Over all there were eight training sets, eight validation sets, and eight testing sets for the model. The training sets spanned eight years each. After the net was run on a forecasting year, the training set window of eight years was moved over cover that year, and the oldest year in the training data set was dropped off. In this way the neural net was kept current as it was forecasted over eight years worth of data. The validation set for the third model was generated randomly from the training set, and the observations were removed from the training set. Summary of the dates used by each of the training and forecasting set are given in table ??.

Data Set 1	Training	Forecasting	Data Set 5	Training	Forecasting
From	Day 7, 1985	Day 328, 1994	From	Day 351, 1988	Day 313, 1998
To	Day 327, 1994	Day 261, 1995	To	Day 310, 1998	Day 308, 1999
Total # of Days	2500 Days	250 Days	Total # of Days	2500 Days	250 Days
Data Set 2	Training	Forecasting	Data Set 6	Training	Forecasting
From	Day 365, 1985	Day 325, 1995	From	Day 347, 1989	Day 309, 1999
To	Day 324, 1995	Day 319, 1996	To	Day 308, 1999	Day 306, 2000
Total # of Days	2500 Days	250 Days	Total # of Days	2500 Days	250 Days
Data Set 3	Training	Forecasting	Data Set 7	Training	Forecasting
From	Day 360, 1986	Day 320, 1996	From	Day 344, 1990	Day 307, 2000
To	Day 319, 1996	Day 315, 1997	To	Day 306, 2000	Day 305, 2001
Total # of Days	2500 Days	250 Days	Total # of Days	2500 Days	250 Days
Data Set 4	Training	Forecasting	Data Set 8	Training	Forecasting
From	Day 356, 1987	Day 316, 1997	From	Day 339, 1991	Day 306, 2001
To	Day 315, 1997	Day 310, 1998	To	Day 305, 2001	Day 304, 2002
Total # of Days	2500 Days	250 Days	Total # of Days	2500 Days	250 Days

Table 4.2: Sample Data Sets for Training and Testing of Model III

4.3 Results

The neural network models were built in MATLAB a software product developed by Mathworks. The program can be installed on any personal computer and is versatile enough to let users make many different types of neural networks. For this study only feed forward networks with varying nodes and functions were tested on the models. As mentioned above, MATLAB restricts the number of observations imported into the software. The number is limited to only 10,000 observation per data set. For this reason the training data set for model I and II was reduced to 250 days.

Many different configurations of feed forward neural networks using back-propagation learning algorithm were used to test the first two models. The best result where given

by a network which had 8 nodes in the input layer, and 8 nodes using tan-sigmoid activation function in the hidden layer. As well as Three or two nodes using the tan-sigmoid activation function in the output layer depending if the network was being trained for Model I and Model II. The results[†] from Model I and II were disappointing and are given in table 4.3.

			1980s	1990s	2000s	3yr Random
R-Square	Model I	1 Month	0.0022	0.0036	0.0006	0.0026
		2 Weeks	0.0442	0.0006	0.0062	0.0474
		1 Week	0.0012	0.0004	0.0593	0.0188
	Model II	1 Week	0.1804	0.0345	0.2777	0.0643
		Basis	0.4301	0.0066	0.0480	0.5695
Correlation	Model I	1 Month	0.0469	-0.0596	0.0235	0.0510
		2 Weeks	-0.2103	-0.0242	-0.0788	0.2176
		1 Week	-0.0347	0.0193	0.2435	-0.1371
	Model II	1 Week	-0.4248	0.1857	-0.5270	0.2536
		Basis	-0.6558	-0.0814	0.2191	-0.7546

Table 4.3: Results of Model I & II

As table 4.3 illustrates none of the models significantly predicted the independent variables. The R^2 of the basis for 1980s and the 3yr Random data set look relatively better than the rest, but an analysis of the correlation suggests that they predicted

[†]Data as well as the full sheets of neural net outputs can be found on the accompanying CD

the prices relatively well in the opposite direction. These results were obtained after significant testing using different variations of back-propagation networks.

Due to the restriction in the data set a smaller model, model III was used. Only the essential variables needed in forecasting the future spot prices of soybeans one month from the current time interval were kept. The decrease in the number of independent variables allowed for data over 8 years to be used to train the neural network. To test model III a new network was built. This new back-propagation network had four input nodes, four nodes using the tan-sigmoid function in the hidden layer, and had one node using tan-sigmoid activation function in the output layer. The neural net performed remarkably better on model III. The performance of the back-propagation neural network predictions of the spot price at the expiration a month before were analyzed. The result of the neural network predictions were compared with those of the futures contracts and a linear regression model which was run on the same data sets as the neural network. There were 21 days on which the comparison was made spanning about eight years from 1994 to 2002. The results[†] of the analysis are given in table 4.4.

The results given in table 4.4 show that over all the back-propagation neural network performed better than the linear regression model, and almost as good as the futures contracts. For years in the 1990s it was seen that the neural network outperformed the futures contracts by a small amount, although in the years after 1999

[†]Data as well as the full sheets of neural net outputs can be found on the accompanying CD

Futures/Actual	2000s	Futures/Actual	1990s	Futures/Actual	Total
<i>R-Square</i>	0.787931175	<i>R-Square</i>	0.961031831	<i>R-Square</i>	0.964347054
<i>Correlation</i>	0.887654874	<i>Correlation</i>	0.98032231	<i>Correlation</i>	0.982011738
ANN Pred/Actual		ANN Pred/Actual		ANN Pred/Actual	
<i>R-Square</i>	0.744567567	<i>R-Square</i>	0.96223419	<i>R-Square</i>	0.962663619
<i>Correlation</i>	0.862883287	<i>Correlation</i>	0.980935365	<i>Correlation</i>	0.981154228
Linear Reg		Linear Reg		Linear Reg	
<i>R-Square</i>	0.752259878	<i>R-Square</i>	0.954498489	<i>R-Square</i>	0.957881286
<i>Correlation</i>	0.867329163	<i>Correlation</i>	0.976984385	<i>Correlation</i>	0.995381988

Table 4.4: Results of Model III

the neural network didn't do as well. After 1999 even the linear model outperformed the neural network.

Based on the predictions produced by the neural network a trading strategy was devised in order to check the efficiency of the soybean futures market. If the predicted spot price was greater than the current spot price then 10 futures contracts were bought, else if the predicted price was less than 10 futures contracts were sold. \$35 dollars were subtracted from the profit upon each transaction accounting for transaction cost involved in making the trade. Since only three transactions were made each year therefore it was assumed that the trader had enough money to carry the transaction out without borrowing. The maximum cost for the trader for the transaction made was \$8,116 dollars. The summary of these transactions and the profits made are given in table 4.5

The results of the trading made on the predictions of the neural network based on model III show that a loss of \$1,482.5 dollars was incurred in the duration of eight

Timeline	Futures	Spot	Spot Forecast	Linear Forecast	Decision	Contract Price @ Exp	Profit
1994:364	550.75	540	546.05	540.70	buy	550	-42.5
1995:116	565	561	562.48	562.90	buy	574.75	62.5
1995:236	597	592.5	591.39	588.69	sell	646.75	-532.5
1995:360	730.5	724	706.31	707.88	sell	731	-40
1996:116	822.5	816	796.08	797.77	sell	796	230
1996:236	803.25	806.25	779.91	780.45	sell	835	-352.5
1996:361	701.75	697.25	695.38	685.38	sell	737	-387.5
1997:114	847	839	813.16	823.44	sell	864.75	-212.5
1997:237	662.75	718	680.25	680.03	sell	756.5	-972.5
1997:358	686.75	676.25	669.71	666.46	sell	656.75	265
1998:114	647.5	640	634.78	640.41	sell	645.5	-15
1998:237	537	541.75	546.24	540.12	buy	532	-85
1998:357	543.5	532.5	522.12	528.22	sell	529	110
1999:113	485.25	473.25	458.09	478.63	sell	468.25	135
1999:237	469.25	456.75	431.51	459.62	sell	481.5	-157.5
1999:351	452.25	438.25	459.13	435.00	buy	485.5	297.5
2000:108	531.5	511	521.34	513.77	buy	551.5	165
2000:238	461.75	465.5	462.10	458.05	sell	483.5	-252.5
2000:350	507.25	492.25	492.95	490.69	buy	479.25	-315
2001:108	434	424.5	443.28	424.98	buy	463	255
2001:228	502.75	515.75	506.27	506.94	sell	471.5	277.5
2001:351	436.75	429.25	431.96	424.69	buy	446	57.5
2002:108	470.25	467.25	476.07	463.36	buy	474.5	7.5
2002:231	569	574.75	546.37	566.19	sell	563.5	20
							-1482.5
							Total

Table 4.5: Results of Buy/Sell Profits based on predictions of Model III

years. Thus it can be stated that the soybean futures market is weak form efficient as per the definition of Fama (1970), for the given data set. Thus the hypothesis that the soybean futures market is not efficient in the weak form is rejected.

4.4 Implications

Considering results from the models used, the question arises that if the market is really efficient or just the neural network structure used, and the method used to train it is not able to predict the future prices with relative accuracy. Thus the relative accuracy condition set by Rausser and Carter (1983) has not been met since overall the neural network did not perform better than the future market even though it did out perform by small amount for Model III. As back-propagation theorem suggests that a neural network does exist that can map the relationship, but finding and training that model is a challenge.

It must also be realized that there were data restriction by the software used to build the ANNs. Due to these data restriction more days worth of training data where not used. Thus the first two models were not able to generalize the relationship between the independent and the dependent variables.

4.5 Limitations & Further Research

This research was limited in time. Therefore other types of networks where not used to forecast using the model developed. There were many problems that arose while using the MATLAB neural network package. The guide book provided with the software was not adequate enough to comprehensively describe all the errors messages that were encountered while building and running neural networks.

A future study in neural network forecasting would benefit from using knowledge collected during this study in effectively building and training neural networks. The

most efficient number of epochs to train the neural network depends on the networks being used, and the problem it is trying to learn. Validation sets are helpful in early stopping, and preventing over fitting of training data by the neural network. Still knowing how to efficiently use these data sets is essential.

There was also constraint on the type of data used in this study. Only the predictors that were available through the DRI database were used since time was limited and could not be waited for data to arrive from the Chicago Board of Trade. Specifically the data of the stocks of soybean oil, soybean meal, and soybeans could have proven to be more helpful in determining the spot price in the future since in the theory section it was shown that the price level of a seasonal commodity was linked to its level of inventory. Level of inventory is also affected by the amount of grain that is exported out of the US. Thus as Rausser and Carter (1983) used the amount of soybean exported out of the US as one of his independent variables used to predict the future spot price of soybeans.

Conclusion

The purpose of this study was to test the efficiency of the soybean futures market in the weak form as presented by Fama (1970) in the efficient market hypothesis through the use of feed-forward neural networks using back-propagation learning algorithm. Neural networks are known for their ability to generalize relationships even if irrelevant data is present. This study was performed using three different back-propagation neural networks, each for the model tested in the study. The results of the third models where relatively better than the other two.

The efficiency of the market was tested through predicting the future spot price of soybeans. The results from the neural network after training on the first two models as mentioned above was not impressive. The third model, a very simple model compared with the first two, showed remarkably better performance in predicting the spot price of soybeans. The remarkably better performance of the third model still didn't match the predictive capabilities of the soybean futures contracts in predicting the future spot price. Therefore this study concludes that given the data, and the neural network used, the soybean futures market are weak form efficient as stated by the efficient market hypothesis.

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